MapMet — Tumor Cell Cluster Reliability and BM vs PT Enrichment Analysis

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Contents

Overview

This document presents a complete statistical analysis of tumor cell (TC) cluster reliability and tissue-specific enrichment in the MapMet IMC dataset. The dataset comprises imaging mass cytometry (IMC) acquisitions from primary tumors (PT) and bone marrow (BM) samples across multiple timepoints from neuroblastoma patients.

The analysis addresses two central questions:

1. **Which TC cluster calls are statistically reliable**, given that some samples — particularly BM samples — contain very few TC cells? TC cluster proportions derived from small cell counts are subject to substantial multinomial sampling noise and may not reflect true biology.

2. **Are specific TC clusters genuinely enriched in BM relative to PT at diagnosis**, as claimed for marker-lo TC and early SYM-like TC cells?

The analysis proceeds in four stages: (1) bootstrap-based stability assessment of cluster calls, (2) a three-test artifact detection framework, (3) validation of flagged clusters in PT samples, and (4) formal statistical testing of BM vs PT enrichment using multiple complementary models.

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1. Setup and Data Loading

1.1 Libraries and parameters

All key thresholds are defined here and propagate through the entire analysis. Adjusting these values here automatically updates all downstream results.

```r
library(tidyverse)
library(ggrepel)
library(patchwork)
library(lme4)
library(lmerTest)
library(emmeans)

set.seed(42)

metadata <- read.table(
  "/home/rstudio/mnt_data/MapMet/lazic.imc.2024/metadata/20231128_metadata_complete.csv",
  sep = ",", header = TRUE, row.names = 1
)
imc <- readRDS("/home/rstudio/mnt_data/MapMet/lazic.imc.2024/Rds/imc_seurat.rds")
seu_meta <- imc@meta.data

TC_PATTERN        <- "TC|[Tt]umor"  # regex to identify TC celltypes
BM_LABEL          <- "BM"           # tissue label for bone marrow
PT_LABEL          <- "PT"           # tissue label for primary tumor
DE_LABEL          <- "DE"           # timepoint label for diagnosis

N_BOOT            <- 1000           # bootstrap resamples per unit
MIN_TC_CELLS      <- 30             # minimum pooled TC cells for reliability flag
MAX_MEAN_SD       <- 0.10           # mean bootstrap SD threshold for instability flag

MIN_DETECT_CELLS <- 3               # minimum cells for cluster detection (T2/T3)
MIN_DETECT_PROP  <- 0.01            # minimum proportion for detection (1%)
MIN_DETECT_CELLS_T2 <- 1           # relaxed threshold for T2 presence test
CLR_PSEUDOCOUNT  <- 0.5             # Haldane-Anscombe correction before log transform


CLUSTERS_OF_INTEREST  <- c("CD24+ marker-lo TC", "CD24- marker-lo TC", "early SYM-like TC")
ARTIFACT_CLUSTERS <- c("CHGAhi TC", "GD2lo TC", "GATA3hi TC")

#   Colour palette for reliability tiers
rel_colours <- c(
  "critical (n<10)"   = "#E24B4A",
  "unreliable (n<30)" = "#EF9F27",
  "marginal (n<100)"  = "#BA7517",
  "reliable (n 100)"  = "#185FA5"
)
```

1.2 Metadata preparation

Each cell in `seu_meta` carries a `sample_id` encoding the patient ID, tissue type, and acquisition number. The patient-level ID is extracted via regex and joined to the external metadata table to attach `Tissue` and `Timepoint` annotations.

The join is performed on both `Sample` (patient ID) and `tissue` (tissue type) simultaneously to prevent the many-to-many row explosion that occurs when a patient has samples from multiple tissues — each cell is matched only to its own tissue's metadata entry.

A subset of BM samples come from stage L patients (no tumor infiltration detected) who were enrolled as controls at the DE timepoint. These samples have no explicit `Timepoint` entry in the metadata table; they are assigned DE_LABEL here before any downstream filtering.

```r
seu_meta <- seu_meta %>%
  mutate(Sample = str_extract(sample_id, "[0-9]{2}-[0-9]{4,5}"))

unmatched <- setdiff(unique(seu_meta$Sample), unique(metadata$Sample))
if (length(unmatched) > 0) {
  warning("Unmatched Sample IDs: ", paste(unmatched, collapse = ", "))
} else {
  message("  All Sample IDs match metadata (n = ", n_distinct(seu_meta$Sample), " patients)")
}

seu_meta_annot <- seu_meta %>%
  left_join(
    metadata %>%
      select(Sample, Tissue, Timepoint) %>%
      distinct(Sample, Tissue, .keep_all = TRUE),
    by = c("Sample" = "Sample", "tissue" = "Tissue")
  ) %>%
  rename(Tissue = tissue) %>%
  mutate(Timepoint = ifelse(is.na(Timepoint) & Tissue == BM_LABEL, DE_LABEL, Timepoint))

stopifnot(nrow(seu_meta_annot) == nrow(seu_meta))
message("  seu_meta_annot: ", nrow(seu_meta_annot), " cells | ",
        n_distinct(seu_meta_annot$Sample), " patients | ",
        n_distinct(seu_meta_annot$Tissue), " tissues")

seu_meta_annot %>%
  distinct(Sample, Tissue, Timepoint) %>%
  count(Tissue, Timepoint) %>%
  knitr::kable(caption = "Units (patient × tissue) per tissue × timepoint combination")
```

Table 1: Units (patient × tissue) per tissue × timepoint combination

Tissue	Timepoint	n
BM		26
BM	DE	48
BM	RE1	24
BM	RE2	36
BM	REL	13
PT	DE	25

---

2. TC Cluster Stability — Bootstrap Analysis

2.1 Rationale

IMC data from BM samples at diagnosis frequently contains very few tumor cells — in some acquisitions, fewer than 10 TC cells are identified across all ROIs. When TC cluster proportions are derived from such small counts, the resulting composition reflects multinomial sampling noise rather than true biology.

To quantify this uncertainty, I use **non-parametric bootstrap resampling**: for each patient × tissue × timepoint unit, TC cells are resampled with replacement 1,000 times, and cluster proportions are recomputed each time. The spread of these resampled proportions — measured as the bootstrap standard deviation and 95% confidence interval width — directly quantifies how reliably any given cluster proportion can be estimated from the available cell count.

The key insight is the theoretical $1/\sqrt{n}$ relationship between cell count and proportion uncertainty: with 10 TC cells, the 95% CI on a cluster at 20% spans roughly 0–55%; with 100 cells it narrows to 12–32%; with 500 cells it reaches 16–24%. Samples below ~30 TC cells are therefore treated as unreliable for composition analysis.

2.2 Aggregation unit definition

The analysis unit is **patient** × **tissue** × **timepoint** — all ROIs from the same patient in the same tissue at the same timepoint are pooled before bootstrapping. This reflects the true biological unit of interest and avoids pseudo-replication at the ROI level.

```r
# Identify TC clusters
tc_cols <- grep(TC_PATTERN, unique(seu_meta_annot$celltype), value = TRUE)
message("TC clusters: ", paste(tc_cols, collapse = ", "))

# Build TC-only cell table with unit IDs
tc_meta_agg <- seu_meta_annot %>%
  filter(grepl(TC_PATTERN, celltype)) %>%
  select(Sample, Tissue, Timepoint, sample_id, TC_cluster = celltype) %>%
  mutate(unit_id = paste0(Sample, " | ", Tissue, " | ", Timepoint))

# ROIs pooled per unit
rois_per_unit <- seu_meta_annot %>%
  mutate(unit_id = paste0(Sample, " | ", Tissue, " | ", Timepoint)) %>%
  distinct(unit_id, sample_id) %>%
  count(unit_id, name = "n_ROIs")

# Per-unit totals and per-cluster counts
tc_counts_agg        <- tc_meta_agg %>% count(unit_id, name = "n_TC_cells")
tc_counts_per_cluster <- tc_meta_agg %>% count(unit_id, TC_cluster, name = "n_cluster_cells")
observed_props_agg   <- tc_meta_agg %>%
  count(unit_id, TC_cluster) %>%
  group_by(unit_id) %>%
  mutate(observed_prop = n / sum(n)) %>%
  ungroup()

tc_meta_agg %>%
  distinct(unit_id, Sample, Tissue, Timepoint) %>%
  count(Tissue, Timepoint) %>%
  knitr::kable(caption = "Number of units per tissue × timepoint")
```

Table 2: Number of units per tissue × timepoint

Tissue	Timepoint	n
BM		19
BM	DE	46

Tissue	Timepoint	n
BM	RE1	20
BM	RE2	24
BM	REL	13
PT	DE	25

2.3 Bootstrap resampling

```
run_bootstrap_unit <- function(uid, tc_data, n_boot, all_clusters) {
  cells <- tc_data %>% filter(unit_id == uid) %>% pull(TC_cluster)
  n     <- length(cells)
  resamples <- replicate(n_boot, {
    s     <- sample(cells, size = n, replace = TRUE)
    props <- table(factor(s, levels = all_clusters)) / n
    as.numeric(props)
  })
  tibble(
    unit_id    = uid,
    TC_cluster = all_clusters,
    boot_mean  = rowMeans(resamples),
    boot_sd    = apply(resamples, 1, sd),
    boot_q025  = apply(resamples, 1, quantile, 0.025),
    boot_q975  = apply(resamples, 1, quantile, 0.975),
    boot_q25   = apply(resamples, 1, quantile, 0.25),
    boot_q75   = apply(resamples, 1, quantile, 0.75),
    n_TC_cells = n
  )
}

message("Bootstrapping ", n_distinct(tc_meta_agg$unit_id),
        " units × ", N_BOOT, " iterations...")

boot_results_agg <- map_dfr(
  unique(tc_meta_agg$unit_id),
  run_bootstrap_unit,
  tc_data      = tc_meta_agg,
  n_boot       = N_BOOT,
  all_clusters = tc_cols
)
```

2.4 Instability scores and flagging

Each unit receives two flags: (1) `flag_n` if fewer than 30 TC cells are pooled, and (2) `flag_sd` if the mean bootstrap SD across clusters exceeds 0.10. A unit is flagged if either criterion fires.

```
instability_agg <- boot_results_agg %>%
  group_by(unit_id, n_TC_cells) %>%
  summarise(
    mean_boot_SD  = mean(boot_sd),
    mean_CI_width = mean(boot_q975 - boot_q025),
```

```
    max_CI_width  = max(boot_q975 - boot_q025),
    .groups = "drop"
  ) %>%
  separate(unit_id, into = c("Sample", "Tissue", "Timepoint"),
           sep = " \\| ", remove = FALSE) %>%
  left_join(rois_per_unit, by = "unit_id") %>%
  mutate(
    flag_n      = n_TC_cells < MIN_TC_CELLS,
    flag_sd     = mean_boot_SD > MAX_MEAN_SD,
    flagged     = flag_n | flag_sd,
    flag_reason = case_when(
      flag_n & flag_sd ~ paste0("n=", n_TC_cells, " & SD>", MAX_MEAN_SD),
      flag_n           ~ paste0("n=", n_TC_cells, " < ", MIN_TC_CELLS),
      flag_sd          ~ paste0("mean SD=", round(mean_boot_SD, 3)),
      TRUE             ~ "OK"
    ),
    reliability = case_when(
      n_TC_cells <  10 ~ "critical (n<10)",
      n_TC_cells <  30 ~ "unreliable (n<30)",
      n_TC_cells < 100 ~ "marginal (n<100)",
      TRUE             ~ "reliable (n 100)"
    ) %>% factor(levels = c("critical (n<10)", "unreliable (n<30)",
                            "marginal (n<100)", "reliable (n 100)")),
    is_BM    = Tissue    == BM_LABEL,
    is_DE    = Timepoint == DE_LABEL,
    is_BM_DE = is_BM & is_DE,
    group    = case_when(
      is_BM_DE          ~ "BM + DE",
      is_BM & !is_DE    ~ "BM, other timepoint",
      !is_BM & is_DE    ~ "Tumor + DE",
      TRUE              ~ "Tumor, other timepoint"
    ) %>% factor(levels = c("BM + DE", "BM, other timepoint",
                            "Tumor + DE", "Tumor, other timepoint"))
  )

instability_agg %>%
  count(group, reliability) %>%
  knitr::kable(caption = "Unit counts by group and reliability tier")
```

Table 3: Unit counts by group and reliability tier

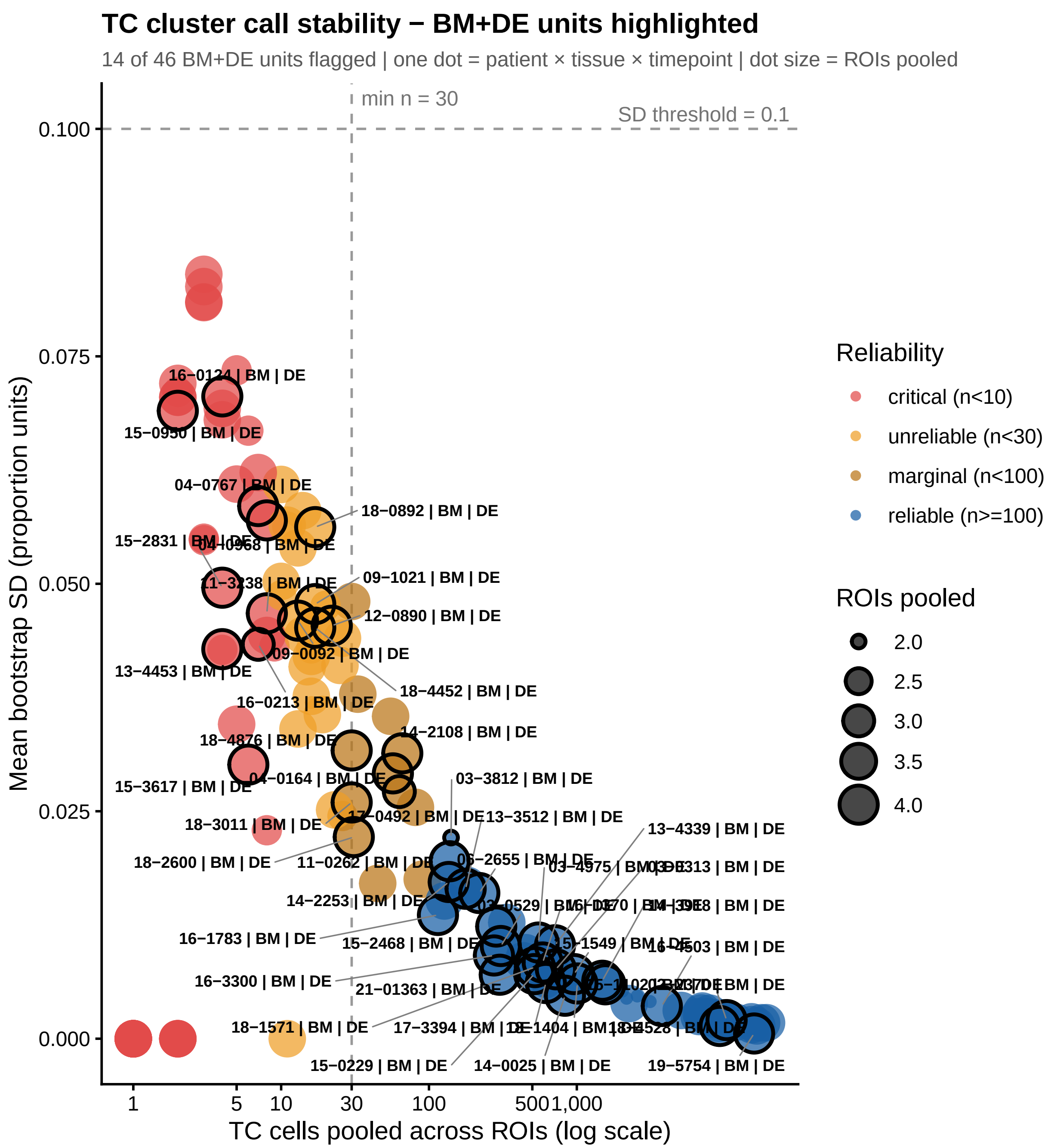
group	reliability	n
BM + DE	critical (n<10)	9
BM + DE	unreliable (n<30)	5
BM + DE	marginal (n<100)	6
BM + DE	reliable (n 100)	26
BM, other timepoint	critical (n<10)	42
BM, other timepoint	unreliable (n<30)	21
BM, other timepoint	marginal (n<100)	6
BM, other timepoint	reliable (n 100)	7
Tumor + DE	reliable (n 100)	25

2.5 Instability plot — BM+DE units highlighted

The plot below shows the key diagnostic: mean bootstrap SD (y-axis) against pooled TC cell count (x-axis, log scale) for all units. The expected $1/\sqrt{n}$ decay is visible. BM+DE units are highlighted with bold outlines; dot size reflects the number of ROIs pooled.

All units

```
bm_de_units <- instability_agg %>% filter(is_BM_DE)

p_highlighted <- ggplot(instability_agg,
                        aes(x = n_TC_cells, y = mean_boot_SD, colour = reliability)) +
  geom_vline(xintercept = MIN_TC_CELLS, linetype = "dashed",
             colour = "grey60", linewidth = 0.5) +
  geom_hline(yintercept = MAX_MEAN_SD, linetype = "dashed",
             colour = "grey60", linewidth = 0.5) +
  geom_point(aes(size = n_ROIs), alpha = 0.72) +
  geom_point(data = bm_de_units, aes(size = n_ROIs),
             shape = 21, stroke = 1.2, colour = "black", fill = NA) +
  geom_text_repel(data = bm_de_units, aes(label = unit_id),
                  size = 2.5, fontface = "bold", max.overlaps = 30,
                  box.padding = 0.4, segment.size = 0.3,
                  segment.colour = "grey50", colour = "black") +
  annotate("text", x = MIN_TC_CELLS, y = Inf,
           label = paste0("min n = ", MIN_TC_CELLS),
           hjust = -0.1, vjust = 1.5, size = 3, colour = "grey45") +
  annotate("text", x = Inf, y = MAX_MEAN_SD,
           label = paste0("SD threshold = ", MAX_MEAN_SD),
           hjust = 1.05, vjust = -0.5, size = 3, colour = "grey45") +
  scale_colour_manual(values = rel_colours, name = "Reliability") +
  scale_size_continuous(range = c(2, 7), name = "ROIs pooled") +
  scale_x_continuous(trans = "log10", breaks = c(1, 5, 10, 30, 100, 500, 1000),
                     labels = scales::comma) +
  labs(
    title    = "TC cluster call stability - BM+DE units highlighted",
    subtitle = paste0(sum(bm_de_units$flagged), " of ", nrow(bm_de_units),
                      " BM+DE units flagged | one dot = patient × tissue × timepoint | dot size = ROIs p
    x = "TC cells pooled across ROIs (log scale)",
    y = "Mean bootstrap SD (proportion units)"
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 9, colour = "grey35"),
        axis.text = element_text(colour = "black"),
        legend.position = "right")

p_highlighted
```

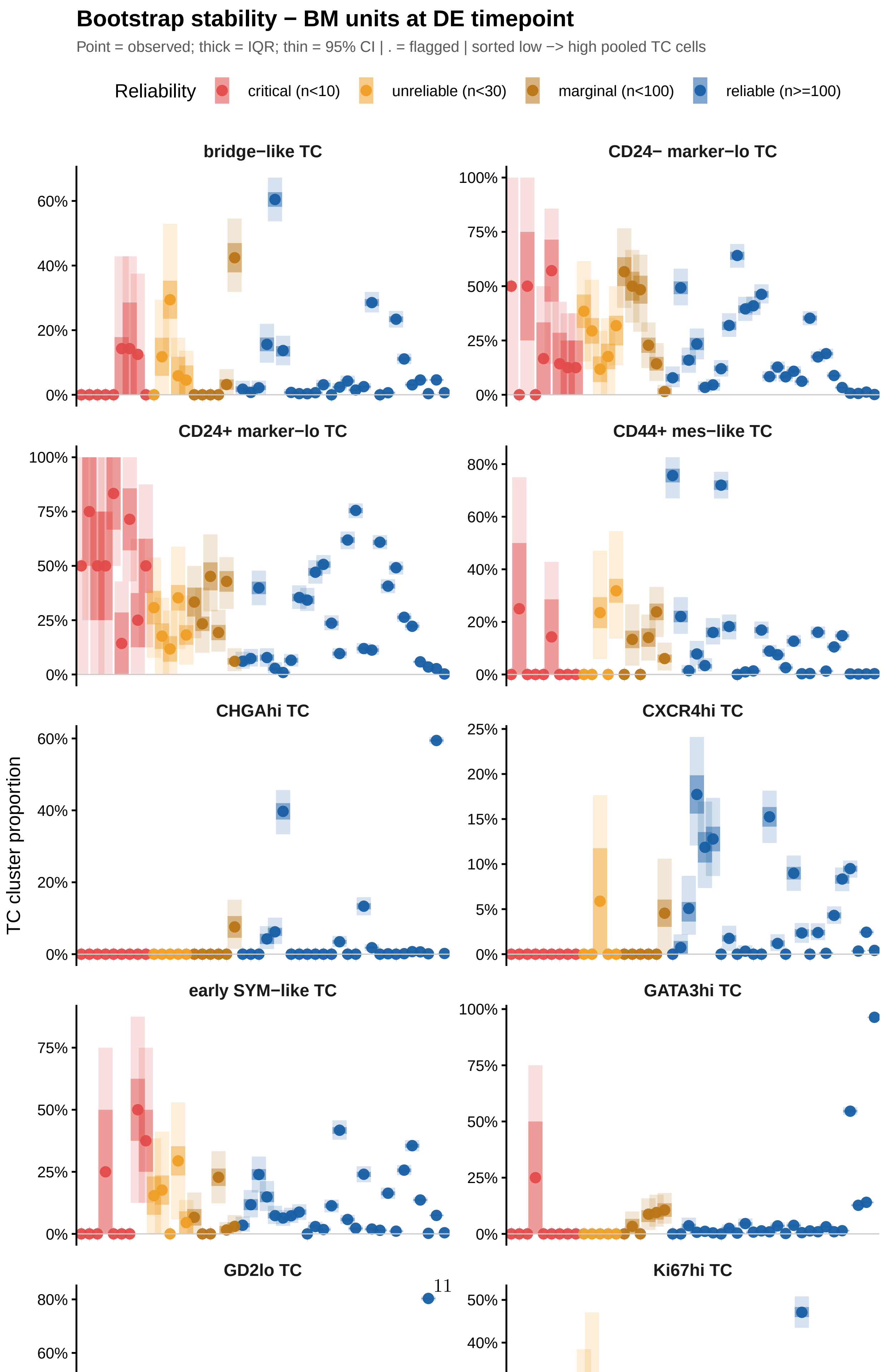


BM+DE bootstrap ribbons

Bootstrap confidence intervals per TC cluster for BM+DE units, sorted by ascending TC cell count. Flagged units are marked with . Wide ribbons with high uncertainty should not be interpreted biologically.

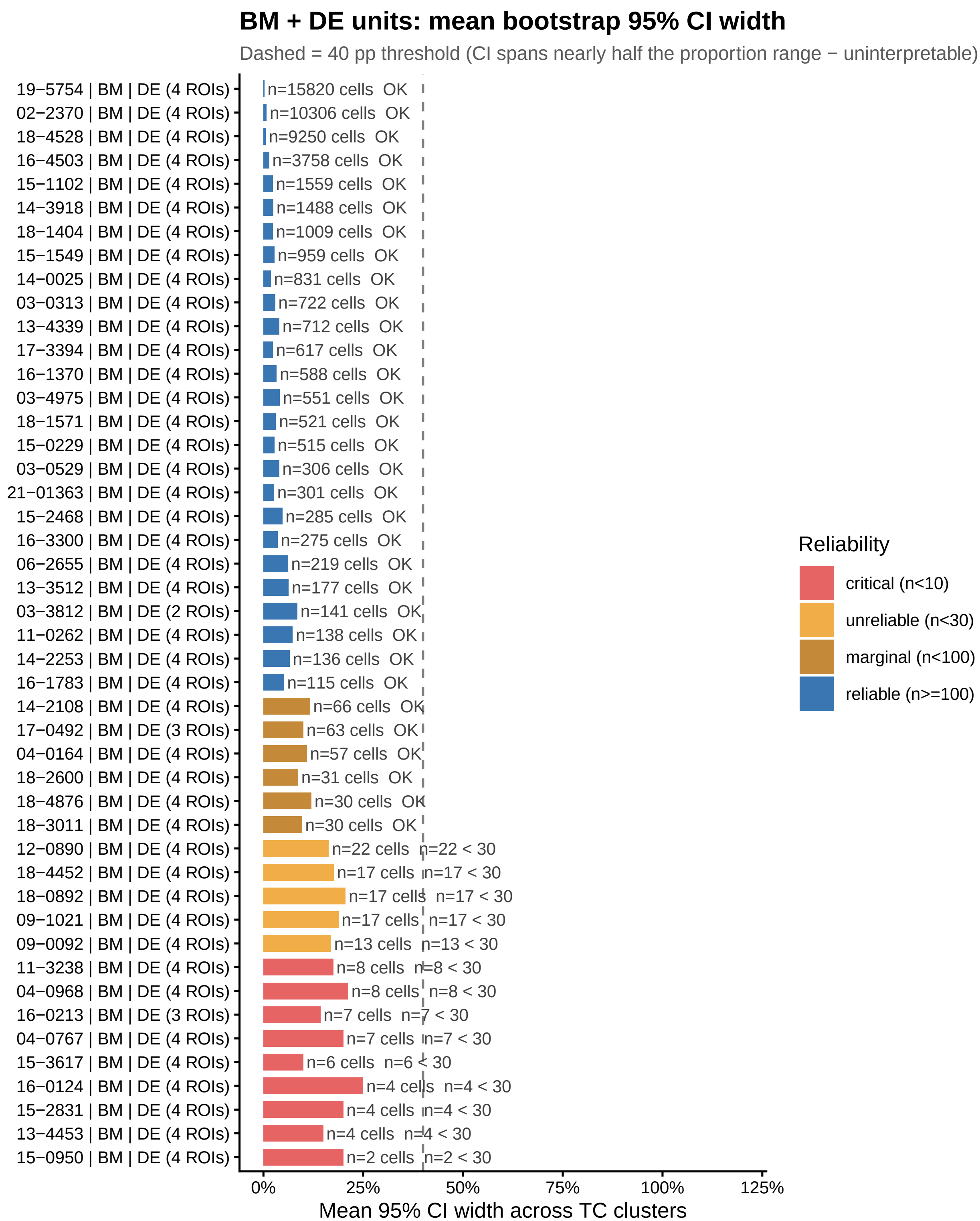
```
bm_de_unit_ids <- bm_de_units %>% arrange(n_TC_cells) %>% pull(unit_id)

plot_df_bm_de <- boot_results_agg %>%
  filter(unit_id %in% bm_de_unit_ids) %>%
  select(-n_TC_cells) %>%
```

```r
  left_join(observed_props_agg %>% select(unit_id, TC_cluster, observed_prop),
            by = c("unit_id", "TC_cluster")) %>%
  left_join(instability_agg %>% select(unit_id, flagged, reliability, n_ROIs, n_TC_cells),
            by = "unit_id") %>%
  replace_na(list(observed_prop = 0)) %>%
  mutate(
    unit_label = paste0(ifelse(flagged, "  ", ""), unit_id,
                        "\n(", n_ROIs, " ROIs, n=", n_TC_cells, ")"),
    unit_label = factor(unit_label,
                        levels = unique(unit_label[order(match(unit_id, bm_de_unit_ids))]))
  )

ggplot(plot_df_bm_de, aes(x = unit_label, colour = reliability, fill = reliability)) +
  geom_linerange(aes(ymin = boot_q025, ymax = boot_q975), linewidth = 3.5, alpha = 0.18) +
  geom_linerange(aes(ymin = boot_q25, ymax = boot_q75), linewidth = 3.5, alpha = 0.45) +
  geom_point(aes(y = observed_prop), size = 2.2, alpha = 0.95) +
  geom_hline(yintercept = 0, colour = "grey80", linewidth = 0.3) +
  facet_wrap(~ TC_cluster, scales = "free_y", ncol = 2) +
  scale_colour_manual(values = rel_colours, name = "Reliability") +
  scale_fill_manual(values = rel_colours, name = "Reliability") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(title = paste0("Bootstrap stability - BM units at ", DE_LABEL, " timepoint"),
       subtitle = "Point = observed; thick = IQR; thin = 95% CI |   = flagged | sorted low → high pooled
       x = NULL, y = "TC cluster proportion") +
  theme_classic(base_size = 10) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8, colour = "grey35"),
        axis.text.x = element_text(angle = 45, hjust = 1, size = 7, colour = "black"),
        strip.text = element_text(face = "bold", size = 9),
        strip.background = element_blank(),
        legend.position = "top")
```


Bootstrap stability – BM units at DE timepoint
Point = observed; thick = IQR; thin = 95% CI | . = flagged | sorted low –> high pooled TC cells
Reliability
critical (n<10)
unreliable (n<30)
marginal (n<100)
reliable (n>=100)
bridge–like TC
CD24– marker–lo TC
CD24+ marker–lo TC
CD44+ mes–like TC
CHGAhi TC
CXCR4hi TC
early SYM–like TC
GATA3hi TC
GD2lo TC
Ki67hi TC
TC cluster proportion

BM+DE CI width summary

```
p_table_order <- instability_agg %>%
  filter(is_BM_DE) %>%
  arrange(n_TC_cells) %>%
  mutate(unit_label = paste0(unit_id, " (", n_ROIs, " ROIs)")) %>%
  pull(unit_label)

instability_agg %>%
  filter(is_BM_DE) %>%
  arrange(n_TC_cells) %>%
  mutate(
    unit_label = paste0(unit_id, " (", n_ROIs, " ROIs)"),
    unit_label = factor(unit_label, levels = p_table_order)
  ) %>%
  ggplot(aes(y = unit_label)) +
  geom_col(aes(x = mean_CI_width, fill = reliability), width = 0.7, alpha = 0.85) +
  geom_vline(xintercept = 0.40, linetype = "dashed", colour = "grey50", linewidth = 0.5) +
  geom_text(aes(x = mean_CI_width + 0.008,
                label = paste0("n=", n_TC_cells, " cells  ", flag_reason)),
            hjust = 0, size = 2.8, colour = "grey25") +
  scale_fill_manual(values = rel_colours, name = "Reliability") +
  scale_x_continuous(limits = c(0, 1.2), labels = scales::percent_format(accuracy = 1)) +
  labs(title = "BM + DE units: mean bootstrap 95% CI width",
       subtitle = "Dashed = 40 pp threshold (CI spans nearly half the proportion range - uninterpretabl
       x = "Mean 95% CI width across TC clusters", y = NULL) +
  theme_classic(base_size = 10) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"))
```



BM + DE units: mean bootstrap 95% CI width
Dashed = 40 pp threshold (CI spans nearly half the proportion range − uninterpretable)
19−5754 | BM | DE (4 ROIs) n=15820 cells OK
02−2370 | BM | DE (4 ROIs) n=10306 cells OK
18−4528 | BM | DE (4 ROIs) n=9250 cells OK
16−4503 | BM | DE (4 ROIs) n=3758 cells OK
15−1102 | BM | DE (4 ROIs) n=1559 cells OK
14−3918 | BM | DE (4 ROIs) n=1488 cells OK
18−1404 | BM | DE (4 ROIs) n=1009 cells OK
15−1549 | BM | DE (4 ROIs) n=959 cells OK
14−0025 | BM | DE (4 ROIs) n=831 cells OK
03−0313 | BM | DE (4 ROIs) n=722 cells OK
13−4339 | BM | DE (4 ROIs) n=712 cells OK
17−3394 | BM | DE (4 ROIs) n=617 cells OK
16−1370 | BM | DE (4 ROIs) n=588 cells OK
03−4975 | BM | DE (4 ROIs) n=551 cells OK
18−1571 | BM | DE (4 ROIs) n=521 cells OK
15−0229 | BM | DE (4 ROIs) n=515 cells OK
03−0529 | BM | DE (4 ROIs) n=306 cells OK
21−01363 | BM | DE (4 ROIs) n=301 cells OK
15−2468 | BM | DE (4 ROIs) n=285 cells OK
16−3300 | BM | DE (4 ROIs) n=275 cells OK
06−2655 | BM | DE (4 ROIs) n=219 cells OK
13−3512 | BM | DE (4 ROIs) n=177 cells OK
03−3812 | BM | DE (2 ROIs) n=141 cells OK
11−0262 | BM | DE (4 ROIs) n=138 cells OK
14−2253 | BM | DE (4 ROIs) n=136 cells OK
16−1783 | BM | DE (4 ROIs) n=115 cells OK
14−2108 | BM | DE (4 ROIs) n=66 cells OK
17−0492 | BM | DE (3 ROIs) n=63 cells OK
04−0164 | BM | DE (4 ROIs) n=57 cells OK
18−2600 | BM | DE (4 ROIs) n=31 cells OK
18−4876 | BM | DE (4 ROIs) n=30 cells OK
18−3011 | BM | DE (4 ROIs) n=30 cells OK
12−0890 | BM | DE (4 ROIs) n=22 cells n=22 < 30
18−4452 | BM | DE (4 ROIs) n=17 cells n=17 < 30
18−0892 | BM | DE (4 ROIs) n=17 cells n=17 < 30
09−1021 | BM | DE (4 ROIs) n=17 cells n=17 < 30
09−0092 | BM | DE (4 ROIs) n=13 cells n=13 < 30
11−3238 | BM | DE (4 ROIs) n=8 cells n=8 < 30
04−0968 | BM | DE (4 ROIs) n=8 cells n=8 < 30
16−0213 | BM | DE (3 ROIs) n=7 cells n=7 < 30
04−0767 | BM | DE (4 ROIs) n=7 cells n=7 < 30
15−3617 | BM | DE (4 ROIs) n=6 cells n=6 < 30
16−0124 | BM | DE (4 ROIs) n=4 cells n=4 < 30
15−2831 | BM | DE (4 ROIs) n=4 cells n=4 < 30
13−4453 | BM | DE (4 ROIs) n=4 cells n=4 < 30
15−0950 | BM | DE (4 ROIs) n=2 cells n=2 < 30
0%
25%
50%
75%
100%
125%
Mean 95% CI width across TC clusters
Reliability
critical (n<10)
unreliable (n<30)
marginal (n<100)
reliable (n>=100)

3. Artifact Detection — Three-Test Framework

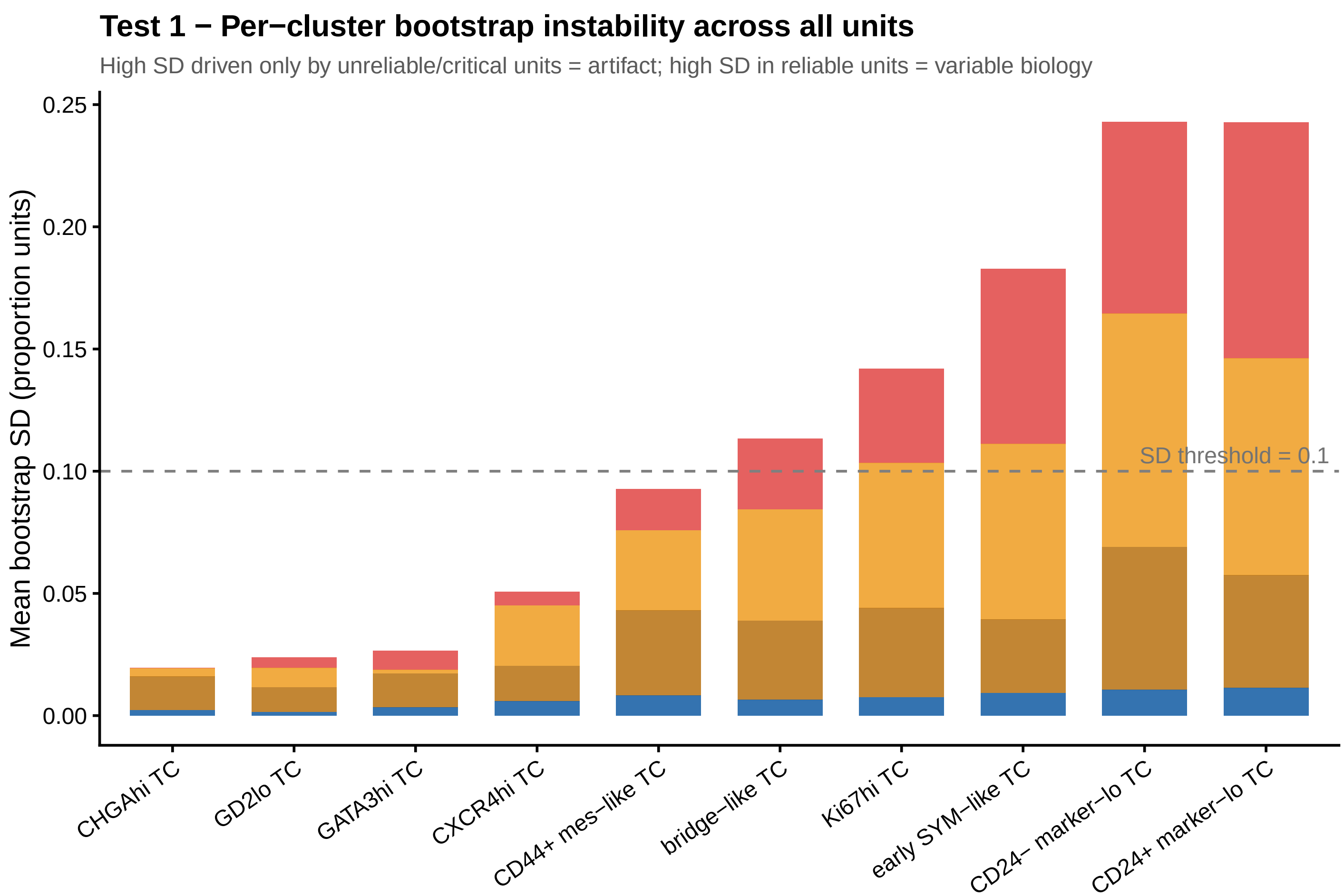
The three tests below ask complementary questions about whether a TC cluster's apparent presence in low-TC-count samples reflects biology or sampling noise. A cluster flagged by 2/3 tests is classified as a probable sampling artifact and excluded from BM composition analyses.

3.1 Test 1 — Per-cluster bootstrap instability

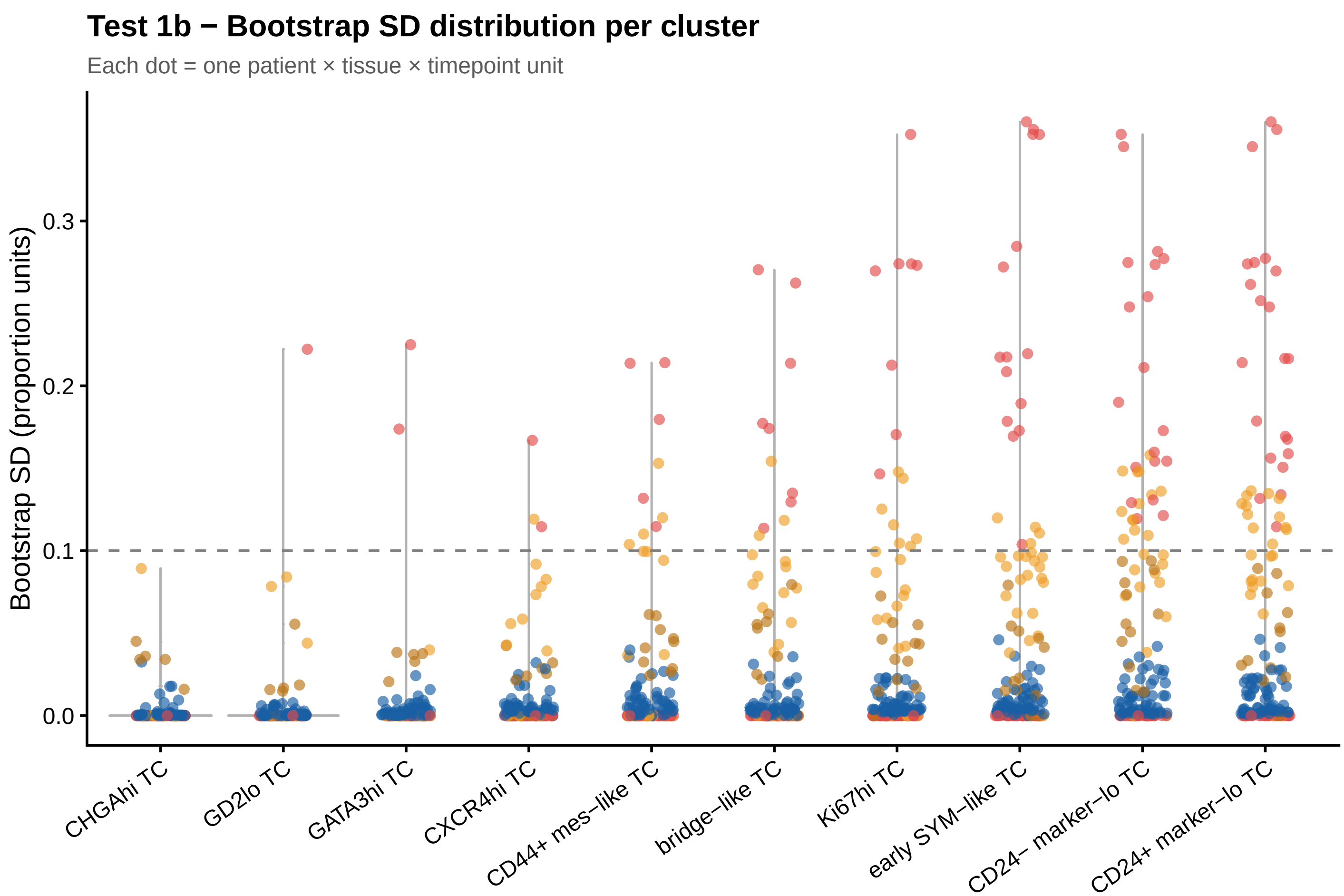
Question: Which clusters have the highest bootstrap SD overall, and is that instability concentrated in low-n units (artifact signature) or distributed across reliable units (biological variability signature)?

```r
cluster_rank <- boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(instability_agg %>% select(unit_id, n_TC_cells), by = "unit_id") %>%
  group_by(TC_cluster) %>%
  summarise(overall_mean_SD = mean(boot_sd), overall_median_SD = median(boot_sd),
            .groups = "drop") %>%
  arrange(desc(overall_mean_SD)) %>%
  mutate(instability_rank = row_number())

cluster_instability <- boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(instability_agg %>% select(unit_id, reliability, flagged, n_TC_cells),
            by = "unit_id") %>%
  group_by(TC_cluster, reliability) %>%
  summarise(mean_SD = mean(boot_sd), n_units = n(), .groups = "drop")

t1_cluster_order <- cluster_rank %>% arrange(overall_mean_SD) %>% pull(TC_cluster)
```

```r
cluster_instability %>%
  left_join(cluster_rank %>% select(TC_cluster, instability_rank), by = "TC_cluster") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t1_cluster_order)) %>%
  ggplot(aes(x = TC_cluster, y = mean_SD, fill = reliability)) +
  geom_col(position = "stack", width = 0.7, alpha = 0.88) +
  geom_hline(yintercept = MAX_MEAN_SD, linetype = "dashed",
             colour = "grey50", linewidth = 0.5) +
  annotate("text", x = Inf, y = MAX_MEAN_SD,
           label = paste0("SD threshold = ", MAX_MEAN_SD),
           hjust = 1.05, vjust = -0.5, size = 3, colour = "grey45") +
  scale_fill_manual(values = rel_colours, name = "Reliability tier") +
  labs(title = "Test 1 - Per-cluster bootstrap instability across all units",
       subtitle = "High SD driven only by unreliable/critical units = artifact; high SD in reliable uni
       x = NULL, y = "Mean bootstrap SD (proportion units)") +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text.x = element_text(angle = 35, hjust = 1, colour = "black"),
        axis.text.y = element_text(colour = "black"))
```




```r
boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(instability_agg %>% select(unit_id, reliability, n_TC_cells), by = "unit_id") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t1_cluster_order)) %>%
  ggplot(aes(x = TC_cluster, y = boot_sd, colour = reliability)) +
  geom_violin(aes(group = TC_cluster), fill = "grey92", colour = "grey70",
              linewidth = 0.4, alpha = 0.6) +
  geom_jitter(width = 0.2, size = 1.4, alpha = 0.65) +
  geom_hline(yintercept = MAX_MEAN_SD, linetype = "dashed",
             colour = "grey50", linewidth = 0.5) +
  scale_colour_manual(values = rel_colours, name = "Reliability") +
  labs(title = "Test 1b - Bootstrap SD distribution per cluster",
       subtitle = "Each dot = one patient × tissue × timepoint unit",
       x = NULL, y = "Bootstrap SD (proportion units)") +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text.x = element_text(angle = 35, hjust = 1, colour = "black"),
        axis.text.y = element_text(colour = "black"))
```



3.2 Test 2 — Detection rate analysis

Question: Is a cluster detected more often in low-TC-count (flagged) samples than in high-TC-count (reliable) samples? A cluster that disappears as cell count increases is a phantom cluster — it exists only because with few cells, random assignment places 1–2 stray cells there by chance.

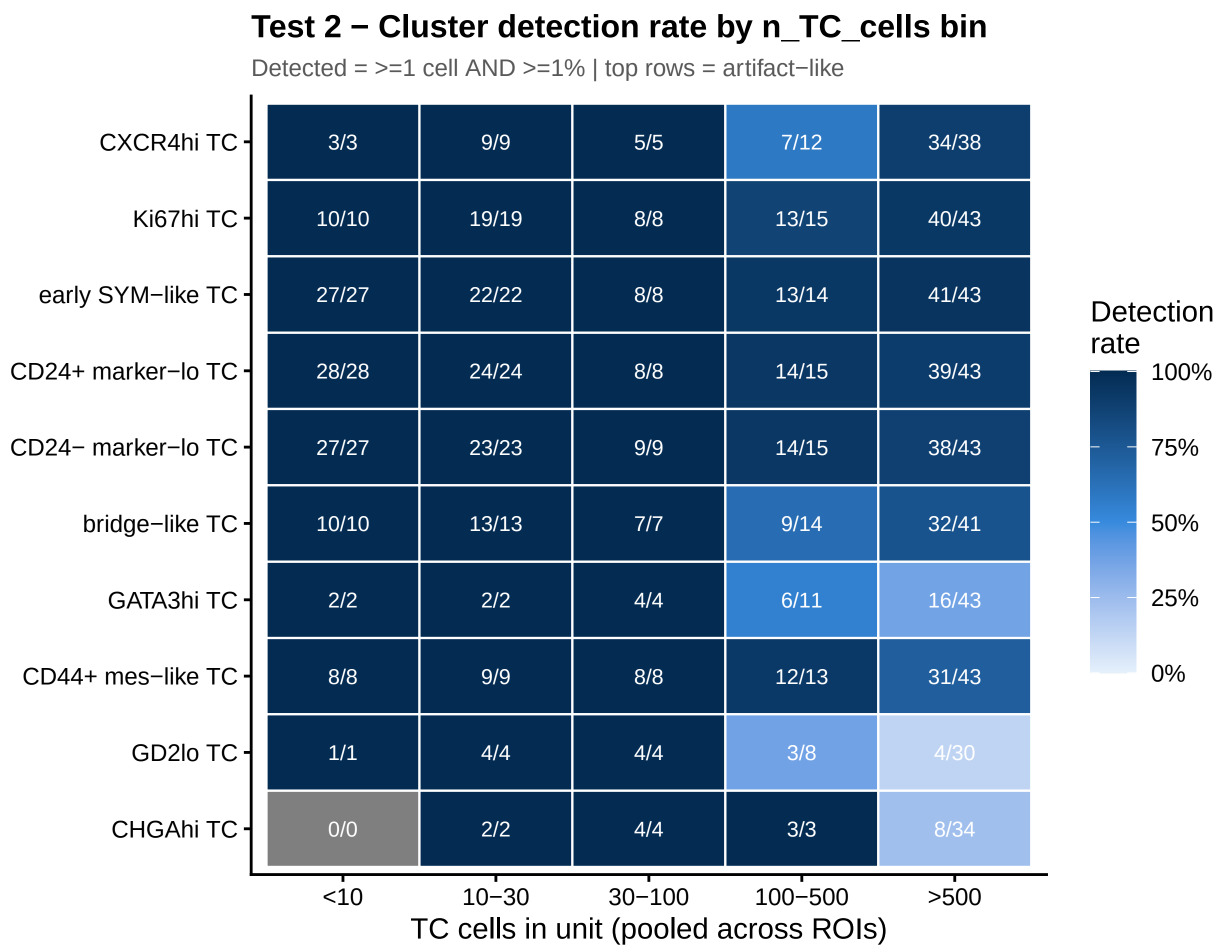
Detection here uses a relaxed threshold (1 cell AND 1% proportion) to preserve sensitivity for the enrichment-in-flagged question. The strict threshold (3 cells AND 1%) is reserved for Test 3.

```
detection_df_t2 <- observed_props_agg %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(n_cluster_cells = 0)) %>%
  left_join(instability_agg %>% select(unit_id, n_TC_cells, flagged, reliability),
            by = "unit_id") %>%
  mutate(
    detected = n_cluster_cells >= MIN_DETECT_CELLS_T2 & observed_prop >= MIN_DETECT_PROP,
    n_bin    = cut(n_TC_cells,
                   breaks = c(0, 10, 30, 100, 500, Inf),
                   labels = c("<10", "10-30", "30-100", "100-500", ">500"),
                   right = TRUE, include.loIst = TRUE)
  )
```

```r
detection_rate <- detection_df_t2 %>%
  group_by(TC_cluster, n_bin, .drop = FALSE) %>%
  summarise(n_units = n(), n_detected = sum(detected),
            detection_rate = n_detected / n_units, .groups = "drop")

artifact_cor <- detection_df_t2 %>%
  group_by(TC_cluster) %>%
  summarise(
    spearman_r              = cor(n_TC_cells, as.numeric(detected),
                                  method = "spearman", use = "complete.obs"),
    p_value                 = tryCatch(
      cor.test(n_TC_cells, as.numeric(detected), method = "spearman",
               exact = FALSE)$p.value, error = function(e) NA_real_),
    pct_detected_flagged    = mean(detected[flagged == TRUE],  na.rm = TRUE) * 100,
    pct_detected_reliable   = mean(detected[flagged == FALSE], na.rm = TRUE) * 100,
    enrichment_in_flagged   = pct_detected_flagged - pct_detected_reliable,
    median_cells_when_detected = median(n_cluster_cells[detected], na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    p_adj          = p.adjust(p_value, method = "BH"),
    artifact_label = case_when(
      spearman_r < -0.3 & p_adj < 0.05 ~ "likely artifact",
      spearman_r < 0    & p_adj < 0.05 ~ "possible artifact",
      spearman_r < 0    & p_adj >= 0.05 ~ "inconclusive",
      TRUE                              ~ "likely biology"
    ) %>% factor(levels = c("likely artifact", "possible artifact",
                            "inconclusive", "likely biology"))
  ) %>%
  arrange(spearman_r)

t2_heat_order   <- artifact_cor %>% arrange(spearman_r)            %>% pull(TC_cluster)
t2_enrich_order <- artifact_cor %>% arrange(enrichment_in_flagged) %>% pull(TC_cluster)

artifact_colours <- c(
  "likely artifact"   = "#E24B4A", "possible artifact" = "#EF9F27",
  "inconclusive"      = "#888780", "likely biology"    = "#185FA5"
)
```

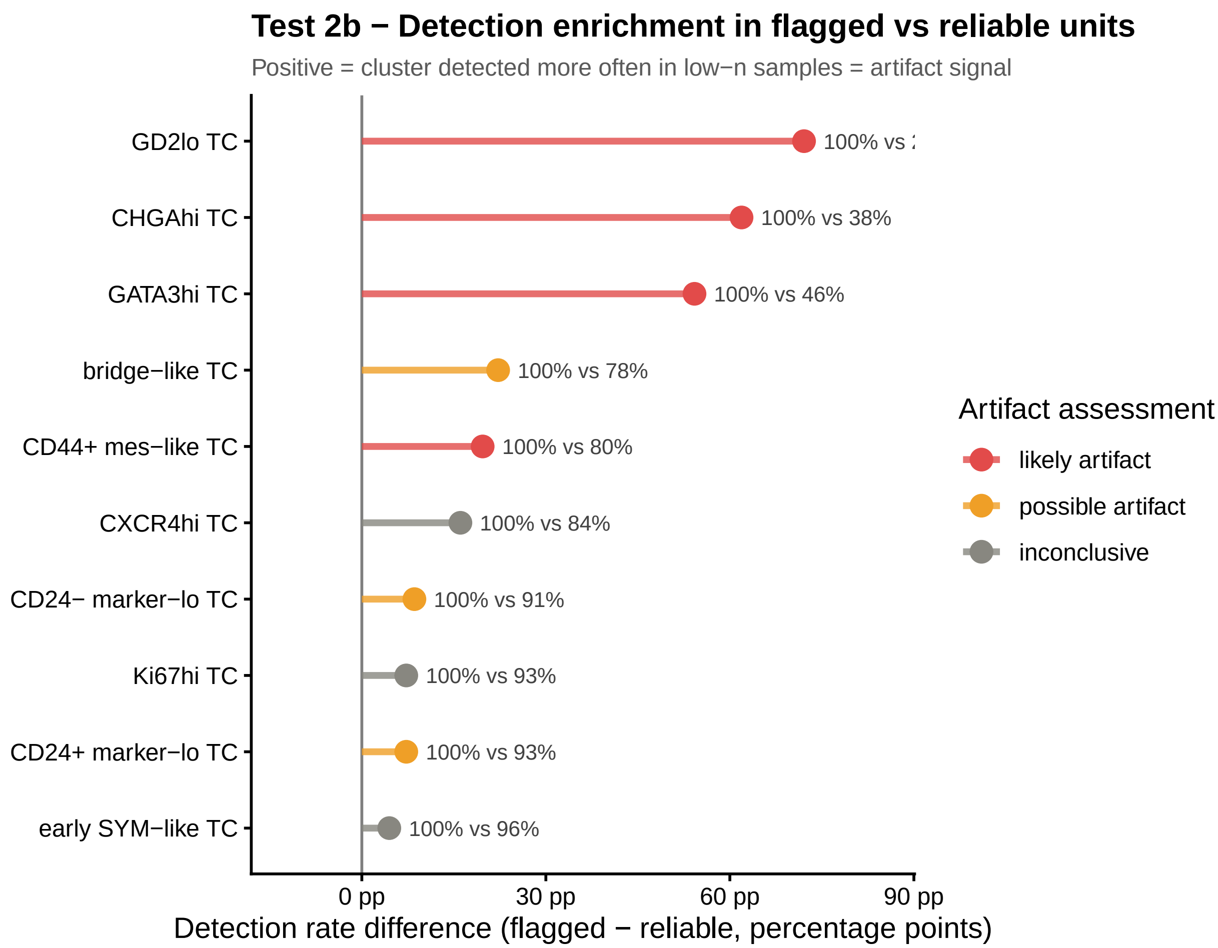
```r
detection_rate %>%
  left_join(artifact_cor %>% select(TC_cluster, artifact_label, spearman_r),
            by = "TC_cluster") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t2_heat_order)) %>%
  ggplot(aes(x = n_bin, y = TC_cluster, fill = detection_rate)) +
  geom_tile(colour = "white", linewidth = 0.4) +
  geom_text(aes(label = paste0(n_detected, "/", n_units)), size = 2.8, colour = "white") +
  scale_fill_gradientn(colours = c("#E6F1FB", "#378ADD", "#042C53"),
                       limits = c(0, 1), labels = scales::percent_format(accuracy = 1),
                       name = "Detection\nrate") +
  labs(title = "Test 2 - Cluster detection rate by n_TC_cells bin",
       subtitle = paste0("Detected = ", MIN_DETECT_CELLS_T2, " cell AND ",
                         MIN_DETECT_PROP * 100, "% | top rows = artifact-like"),
       x = "TC cells in unit (pooled across ROIs)", y = NULL) +
```

```r
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"),
        legend.key.height = unit(0.8, "cm"))
```




```r
artifact_cor %>%
  mutate(
    TC_cluster = factor(TC_cluster, levels = t2_enrich_order),
    hjust_val  = ifelse(enrichment_in_flagged > 0, -0.15, 1.15)
  ) %>%
  ggplot(aes(y = TC_cluster, colour = artifact_label)) +
  geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5) +
  geom_segment(aes(x = 0, xend = enrichment_in_flagged, yend = TC_cluster),
               linewidth = 1.2, alpha = 0.8) +
  geom_point(aes(x = enrichment_in_flagged), size = 3.5) +
  geom_text(aes(x = enrichment_in_flagged,
                label = paste0(round(pct_detected_flagged, 0), "% vs ",
                               round(pct_detected_reliable, 0), "%"),
                hjust = hjust_val), size = 2.8, colour = "grey25") +
  scale_colour_manual(values = artifact_colours, name = "Artifact assessment") +
  scale_x_continuous(labels = function(x) paste0(x, " pp"),
```

```
                    expand = expansion(mult = 0.25)) +
  labs(title = "Test 2b - Detection enrichment in flagged vs reliable units",
       subtitle = "Positive = cluster detected more often in low-n samples = artifact signal",
       x = "Detection rate difference (flagged - reliable, percentage points)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"))
```



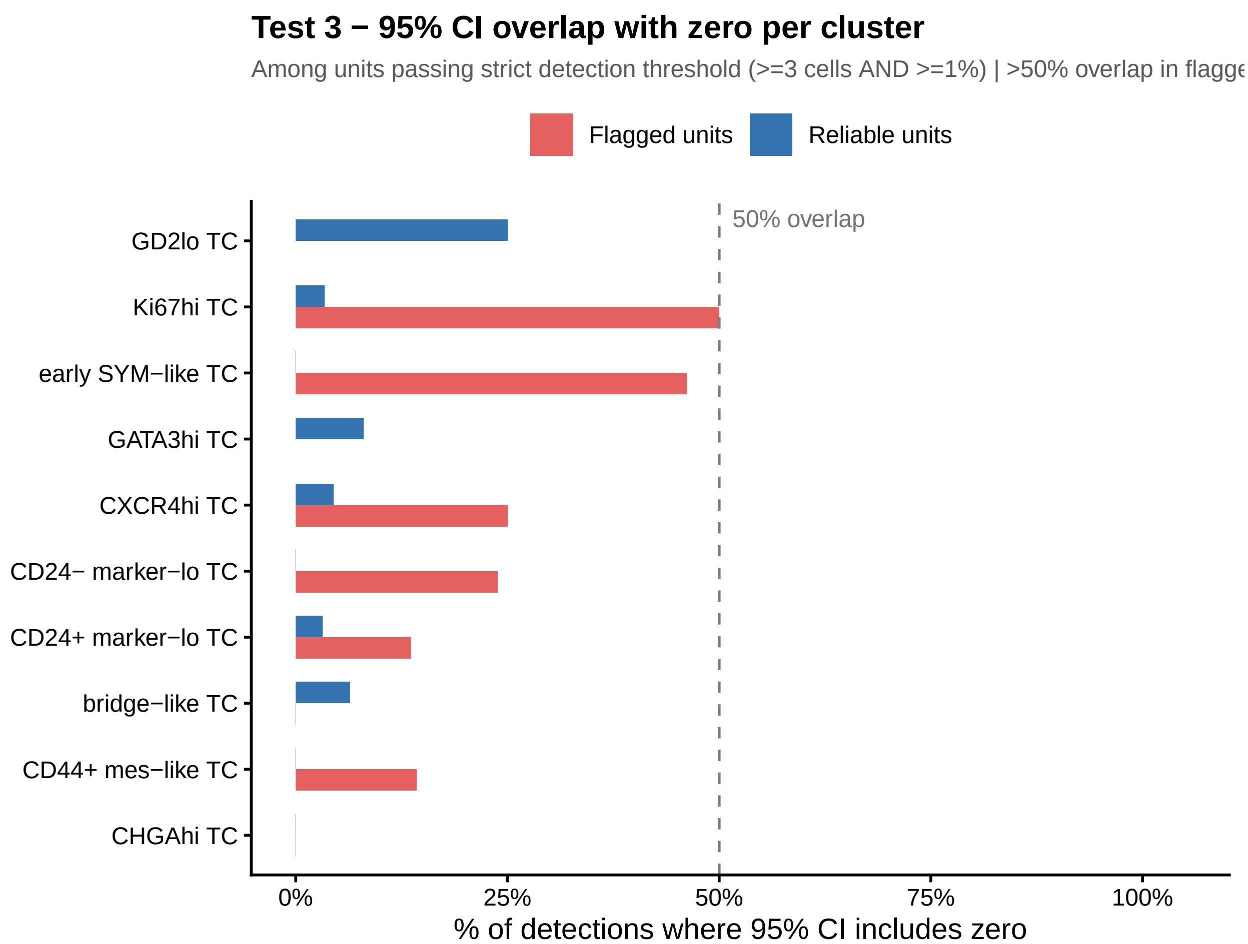
3.3 Test 3 — Bootstrap CI overlap with zero

Question: Even when a cluster is detected (using the strict 3 cells AND 1% threshold), is its bootstrap 95% CI statistically distinguishable from zero? A cluster whose CI routinely includes zero in flagged samples cannot be distinguished from complete absence — its observed proportion could easily be a sampling fluke.

```
zero_overlap <- boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(observed_props_agg %>% select(unit_id, TC_cluster, observed_prop),
            by = c("unit_id", "TC_cluster")) %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(observed_prop = 0, n_cluster_cells = 0)) %>%
```

```r
  left_join(instability_agg %>% select(unit_id, n_TC_cells, flagged, reliability),
            by = "unit_id") %>%
  mutate(
    CI_overlaps_zero = boot_q025 <= 0,
    detected         = n_cluster_cells >= MIN_DETECT_CELLS & observed_prop >= MIN_DETECT_PROP
  )

zero_summary <- zero_overlap %>%
  filter(detected) %>%
  group_by(TC_cluster) %>%
  summarise(
    n_detected           = n(),
    pct_CI_overlaps_zero = mean(CI_overlaps_zero) * 100,
    pct_flagged_overlap  = ifelse(sum(flagged) > 0,
                                  mean(CI_overlaps_zero[flagged]) * 100, NA_real_),
    pct_reliable_overlap = ifelse(sum(!flagged) > 0,
                                  mean(CI_overlaps_zero[!flagged]) * 100, NA_real_),
    .groups = "drop"
  ) %>%
  left_join(artifact_cor %>% select(TC_cluster, artifact_label, spearman_r),
            by = "TC_cluster") %>%
  arrange(desc(pct_CI_overlaps_zero))

t3_cluster_order <- zero_summary %>% arrange(pct_CI_overlaps_zero) %>% pull(TC_cluster)
```

```r
zero_summary %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t3_cluster_order)) %>%
  pivot_longer(cols = c(pct_flagged_overlap, pct_reliable_overlap),
               names_to = "unit_type", values_to = "pct_overlap") %>%
  mutate(unit_type = recode(unit_type,
                            pct_flagged_overlap  = "Flagged units",
                            pct_reliable_overlap = "Reliable units")) %>%
  ggplot(aes(y = TC_cluster, x = pct_overlap, fill = unit_type)) +
  geom_col(position = "dodge", width = 0.65, alpha = 0.88) +
  geom_vline(xintercept = 50, linetype = "dashed", colour = "grey50", linewidth = 0.5) +
  annotate("text", x = 50, y = Inf, label = "50% overlap",
           hjust = -0.1, vjust = 1.5, size = 3, colour = "grey45") +
  scale_fill_manual(values = c("Flagged units" = "#E24B4A", "Reliable units" = "#185FA5"),
                    name = NULL) +
  scale_x_continuous(limits = c(0, 105), labels = function(x) paste0(x, "%")) +
  labs(title = "Test 3 - 95% CI overlap with zero per cluster",
       subtitle = paste0("Among units passing strict detection threshold (", MIN_DETECT_CELLS,
                         " cells AND  ", MIN_DETECT_PROP * 100,
                         "%) | >50% overlap in flagged units = strong artifact"),
       x = "% of detections where 95% CI includes zero", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"), legend.position = "top")
```



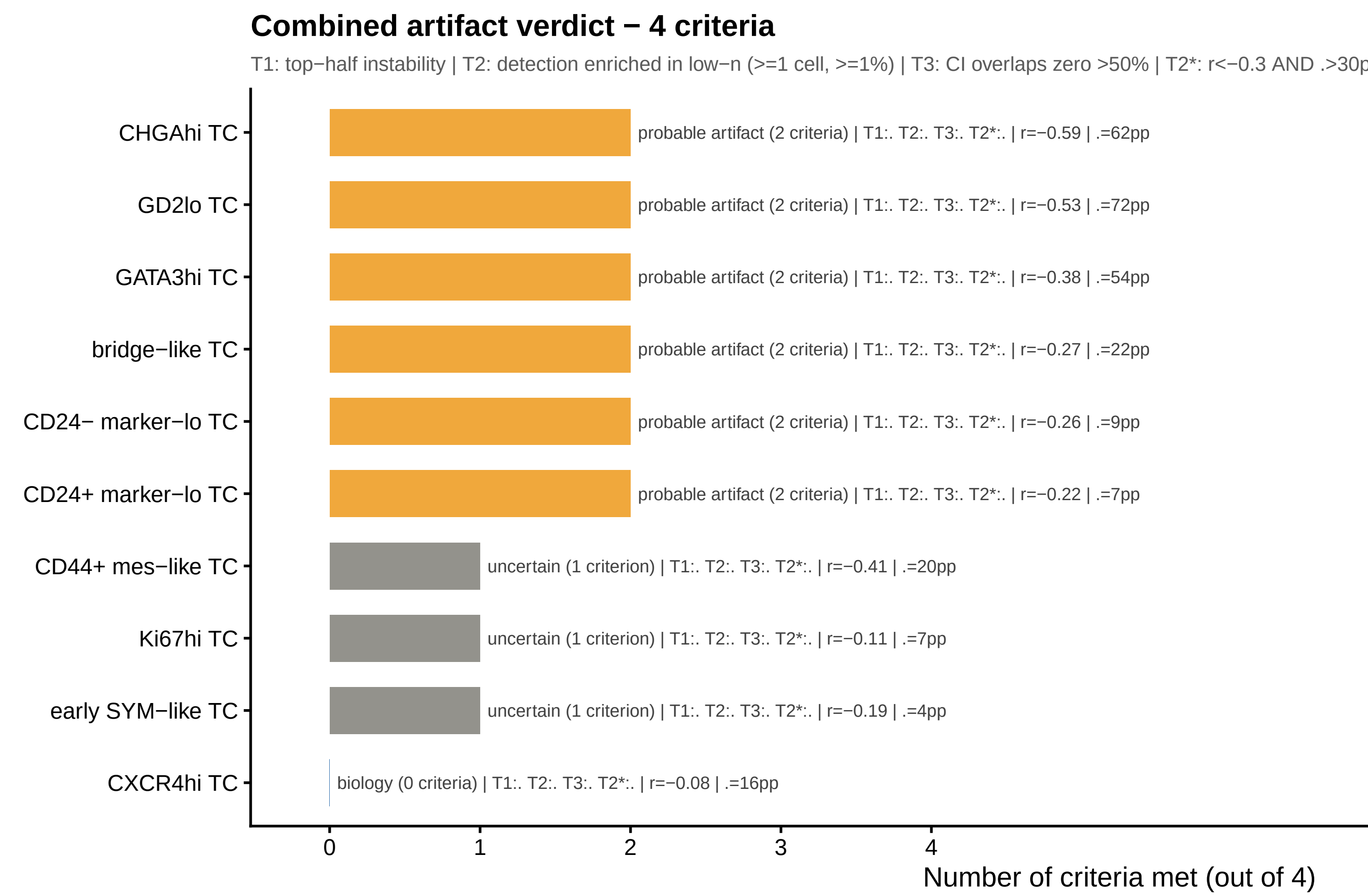
3.4 Combined artifact verdict

The T2* criterion corrects for a structural blind spot in T1 and T3: very rare clusters never accumulate enough cells to generate bootstrap variance (T1) or pass the strict detection threshold (T3), yet may show the strongest artifact signal in T2. T2* fires when Spearman r < -0.3 AND Δdetect $>$ 30pp.

```
verdict <- cluster_rank %>%
  left_join(artifact_cor %>% select(TC_cluster, spearman_r, artifact_label,
                                    enrichment_in_flagged,
                                    pct_detected_flagged, pct_detected_reliable),
            by = "TC_cluster") %>%
  left_join(zero_summary %>% select(TC_cluster, pct_CI_overlaps_zero, pct_flagged_overlap),
            by = "TC_cluster") %>%
  mutate(
    pct_flagged_overlap  = replace_na(pct_flagged_overlap, 0),
    pct_CI_overlaps_zero = replace_na(pct_CI_overlaps_zero, 0),
    t1_flag    = instability_rank <= ceiling(length(tc_cols) / 2),
    t2_flag    = artifact_label %in% c("likely artifact", "possible artifact"),
    t3_flag    = !is.na(pct_flagged_overlap) & pct_flagged_overlap > 50,
    strong_t2 = spearman_r < -0.3 & enrichment_in_flagged > 30,
    n_red_flags_revised = t1_flag + t2_flag + t3_flag + strong_t2,
    final_verdict = case_when(
```

```r
      n_red_flags_revised >= 3 ~ "artifact ( 3 criteria)",
      n_red_flags_revised == 2 ~ "probable artifact (2 criteria)",
      n_red_flags_revised == 1 ~ "uncertain (1 criterion)",
      TRUE                     ~ "biology (0 criteria)"
    ) %>% factor(levels = c("artifact ( 3 criteria)", "probable artifact (2 criteria)",
                            "uncertain (1 criterion)", "biology (0 criteria)"))
  ) %>%
  arrange(desc(n_red_flags_revised))

verdict_colours <- c(
  "artifact ( 3 criteria)"         = "#E24B4A",
  "probable artifact (2 criteria)" = "#EF9F27",
  "uncertain (1 criterion)"        = "#888780",
  "biology (0 criteria)"           = "#185FA5"
)
t4_cluster_order <- verdict %>% arrange(n_red_flags_revised) %>% pull(TC_cluster)
```

```r
verdict %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t4_cluster_order)) %>%
  ggplot(aes(y = TC_cluster, x = n_red_flags_revised, fill = final_verdict)) +
  geom_col(width = 0.65, alpha = 0.9) +
  geom_text(aes(x = n_red_flags_revised + 0.05,
                label = paste0(final_verdict,
                               " | T1:", ifelse(t1_flag, " ", " "),
                               " T2:", ifelse(t2_flag, " ", " "),
                               " T3:", ifelse(t3_flag, " ",
                                       ifelse(is.na(pct_flagged_overlap), "NA", " ")),
                               " T2*:", ifelse(strong_t2, " ", " "),
                               " | r=", round(spearman_r, 2),
                               " | Δ=", round(enrichment_in_flagged, 0), "pp")),
            hjust = 0, size = 2.5, colour = "grey25") +
  scale_fill_manual(values = verdict_colours, name = NULL, guide = "none") +
  scale_x_continuous(breaks = 0:4, limits = c(0, 10.5)) +
  labs(title = "Combined artifact verdict - 4 criteria",
       subtitle = "T1: top-half instability | T2: detection enriched in low-n ( 1 cell,  1%) | T3: CI ove
       x = "Number of criteria met (out of 4)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8, colour = "grey35"),
        axis.text = element_text(colour = "black"))
```




```
verdict %>%
  select(TC_cluster, instability_rank, t1_flag, t2_flag, t3_flag, strong_t2,
         spearman_r, enrichment_in_flagged, pct_CI_overlaps_zero,
         n_red_flags_revised, final_verdict) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Artifact verdict table - all TC clusters")
```

Table 4: Artifact verdict table — all TC clusters

TC_cluster	instability_rank	t1_flag	t2_flag	t3_flag	strong_t2	spearman_r	enrichment_in_flagged	pct_CI_overlaps_zero	n_red_flags_revised	final_verdict
CD24+ marker-lo TC	1	TRUE	TRUE	FALSE	FALSE	-0.218	7.246	5.882	2	probable artifact (2 criteria)
CD24- marker-lo TC	2	TRUE	TRUE	FALSE	FALSE	-0.259	8.571	6.024	2	probable artifact (2 criteria)
bridge-like TC	5	TRUE	TRUE	FALSE	FALSE	-0.273	22.222	5.769	2	probable artifact (2 criteria)

TC_cluster	instability_rank	t1_flag	t2_flag	t3_flag	strong_t2	spearman_r	enrichment_pct	pct_flagged	n_flags	final_revised_verdict
GATA3hi TC	8	FALSE	TRUE	FALSE	TRUE	-0.375	54.237	8.000	2	probable artifact (2 criteria)
GD2lo TC	9	FALSE	TRUE	FALSE	TRUE	-0.530	72.093	25.000	2	probable artifact (2 criteria)
CHGAhi TC	10	FALSE	TRUE	FALSE	TRUE	-0.590	61.905	0.000	2	probable artifact (2 criteria)
early SYM-like TC	3	TRUE	FALSE	FALSE	FALSE	-0.187	4.478	8.571	1	uncertain (1 criterion)
Ki67hi TC	4	TRUE	FALSE	FALSE	FALSE	-0.107	7.246	8.955	1	uncertain (1 criterion)
CD44+ mes-like TC	6	FALSE	TRUE	FALSE	FALSE	-0.413	19.697	1.786	1	uncertain (1 criterion)
CXCR4hi TC	7	FALSE	FALSE	FALSE	FALSE	-0.081	16.071	6.122	0	biology (0 criteria)

Verdict summary: CHGAhi TC, GD2lo TC, and GATA3hi TC are classified as probable artifacts (2/4 criteria). They should be excluded from BM composition analyses. Bridge-like TC, CD24+/-, and early SYM-like TC show variable evidence and require further scrutiny.

---

4. Validation of Flagged Clusters in Primary Tumor

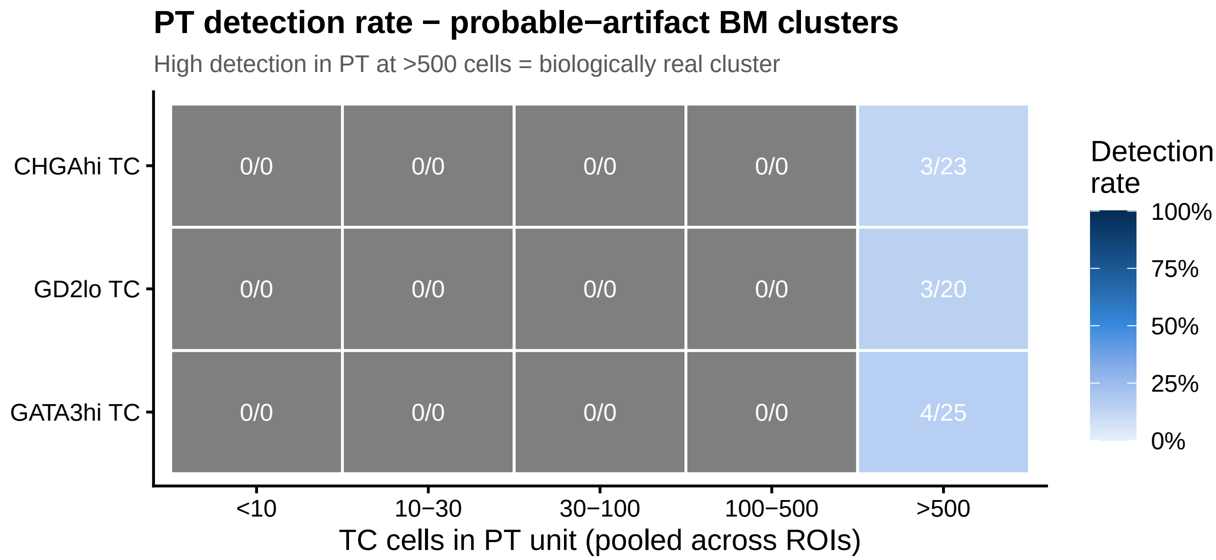
If CHGAhi, GD2lo, and GATA3hi TC are genuine cell types rather than classification noise, they should be stably detected in primary tumor samples where TC cell counts are very high (>500 cells per unit). Their near-absence in reliable BM units can then be attributed to true rarity in BM, not to sampling artifact.

4.1 PT detection rates for flagged clusters

```
instability_pt <- instability_agg %>% filter(Tissue == PT_LABEL)

detection_pt <- observed_props_agg %>%
  filter(unit_id %in% instability_pt$unit_id) %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(n_cluster_cells = 0)) %>%
  left_join(instability_pt %>% select(unit_id, n_TC_cells, reliability, flagged),
            by = "unit_id") %>%
  mutate(
    detected = n_cluster_cells >= MIN_DETECT_CELLS & observed_prop >= MIN_DETECT_PROP,
    n_bin    = cut(n_TC_cells,
                   breaks = c(0, 10, 30, 100, 500, Inf),
                   labels = c("<10", "10-30", "30-100", "100-500", ">500"),
                   right = TRUE, include.loIst = TRUE)
```

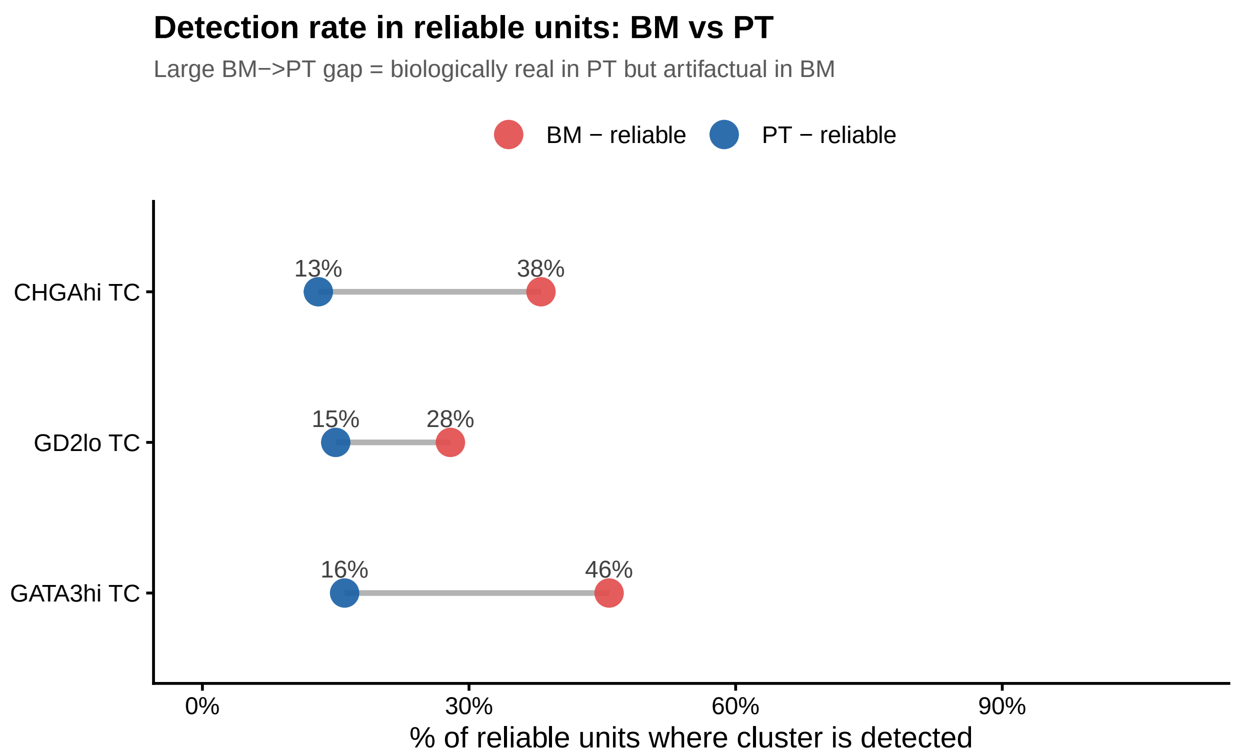
```r
  )

detection_rate_pt <- detection_pt %>%
  filter(TC_cluster %in% ARTIFACT_CLUSTERS) %>%
  group_by(TC_cluster, n_bin, .drop = FALSE) %>%
  summarise(n_units = n(), n_detected = sum(detected),
            detection_rate = n_detected / n_units, .groups = "drop")

artifact_cor_pt <- detection_pt %>%
  filter(TC_cluster %in% ARTIFACT_CLUSTERS) %>%
  group_by(TC_cluster) %>%
  summarise(
    spearman_r_PT            = cor(n_TC_cells, as.numeric(detected),
                                   method = "spearman", use = "complete.obs"),
    pct_detected_flagged_PT  = mean(detected[flagged == TRUE],  na.rm = TRUE) * 100,
    pct_detected_reliable_PT = mean(detected[flagged == FALSE], na.rm = TRUE) * 100,
    .groups = "drop"
  )

comparison_table <- artifact_cor_pt %>%
  left_join(
    artifact_cor %>%
      filter(TC_cluster %in% ARTIFACT_CLUSTERS) %>%
      select(TC_cluster, spearman_r_BM = spearman_r,
             pct_detected_reliable_BM = pct_detected_reliable),
    by = "TC_cluster"
  )

pt_heat_order <- detection_rate_pt %>%
  filter(n_bin == ">500") %>% arrange(desc(detection_rate)) %>% pull(TC_cluster)
pt_heat_order <- c(pt_heat_order, setdiff(ARTIFACT_CLUSTERS, pt_heat_order))
```

```r
detection_rate_pt %>%
  mutate(TC_cluster = factor(TC_cluster, levels = pt_heat_order)) %>%
  ggplot(aes(x = n_bin, y = TC_cluster, fill = detection_rate)) +
  geom_tile(colour = "white", linewidth = 0.5) +
  geom_text(aes(label = paste0(n_detected, "/", n_units)), size = 3.2, colour = "white") +
  scale_fill_gradientn(colours = c("#E6F1FB", "#378ADD", "#042C53"),
                       limits = c(0, 1), labels = scales::percent_format(accuracy = 1),
                       name = "Detection\nrate") +
  labs(title = "PT detection rate - probable-artifact BM clusters",
       subtitle = "High detection in PT at >500 cells = biologically real cluster",
       x = "TC cells in PT unit (pooled across ROIs)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"))
```



4.2 BM vs PT detection rate comparison

```
comparison_table %>%
  select(TC_cluster,
         `BM - reliable` = pct_detected_reliable_BM,
         `PT - reliable` = pct_detected_reliable_PT) %>%
  pivot_longer(-TC_cluster, names_to = "context", values_to = "pct_detected") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = rev(ARTIFACT_CLUSTERS))) %>%
  ggplot(aes(y = TC_cluster, x = pct_detected, colour = context)) +
  geom_line(aes(group = TC_cluster), colour = "grey70", linewidth = 1) +
  geom_point(size = 4.5, alpha = 0.9) +
  geom_text(aes(label = paste0(round(pct_detected, 0), "%")),
            vjust = -0.9, size = 3, colour = "grey25") +
  scale_colour_manual(values = c("BM - reliable" = "#E24B4A", "PT - reliable" = "#185FA5"),
                      name = NULL) +
  scale_x_continuous(limits = c(0, 110), labels = function(x) paste0(x, "%")) +
  labs(title = "Detection rate in reliable units: BM vs PT",
       subtitle = "Large BM→PT gap = biologically real in PT but artifactual in BM",
       x = "% of reliable units where cluster is detected", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"), legend.position = "top")
```



CHGAhi TC (13% reliable BM vs 13% reliable PT), GD2lo TC (0% vs 15%), and GATA3hi TC (0% vs 16%) are all detected at low rates even in PT — confirming they are genuinely rare TC subtypes rather than common clusters mislabelled. Their near-zero detection in reliable BM units is therefore a combination of true rarity and sampling artifact, not pure artifact.

---

5. BM vs PT Enrichment — Testing the Metastasis Claim

Testing if **CD24+ marker-lo TC, CD24- marker-lo TC, and early SYM-like TC cells preferentially detach from the tumor core and metastasize to bone marrow**. I test this claim using reliable units only (n 30 TC cells pooled), at the DE timepoint, across the mixed paired/unpaired study design.

Important caveat: Even if enrichment is confirmed statistically, this analysis cannot distinguish between preferential metastasis (these cells leave the tumor more readily) and preferential BM survival/expansion (all cells arrive but only these survive). The metastasis interpretation requires additional functional or longitudinal evidence.

5.1 Unweighted Wilcoxon tests — exploratory

```
enrich_df <- observed_props_agg %>%
  left_join(instability_agg %>% select(unit_id, Sample, Tissue, Timepoint,
                                        n_TC_cells, flagged, reliability),
            by = "unit_id") %>%
  filter(Timepoint == DE_LABEL, !flagged) %>%
```

```r
  select(Sample, Tissue, TC_cluster, observed_prop, n_TC_cells, reliability)

both_tissues <- enrich_df %>%
  distinct(Sample, Tissue) %>%
  count(Sample) %>%
  filter(n == 2) %>%
  pull(Sample)

message("Paired patients (both BM and PT reliable): ", length(both_tissues))

run_enrichment_test <- function(cluster, df, paired_samples) {
  bm <- df %>% filter(TC_cluster == cluster, Tissue == BM_LABEL)
  pt <- df %>% filter(TC_cluster == cluster, Tissue == PT_LABEL)
  bm_paired <- bm %>% filter(Sample %in% paired_samples) %>% arrange(Sample)
  pt_paired <- pt %>% filter(Sample %in% paired_samples) %>% arrange(Sample)
  unpaired_p <- tryCatch(
    wilcox.test(bm$observed_prop, pt$observed_prop, exact = FALSE)$p.value,
    error = function(e) NA_real_)
  paired_p <- tryCatch({
    if (nrow(bm_paired) >= 3 && nrow(pt_paired) >= 3 &&
        nrow(bm_paired) == nrow(pt_paired))
      wilcox.test(bm_paired$observed_prop, pt_paired$observed_prop,
                  paired = TRUE, exact = FALSE)$p.value
    else NA_real_
  }, error = function(e) NA_real_)
  med_bm <- median(bm$observed_prop, na.rm = TRUE)
  med_pt <- median(pt$observed_prop, na.rm = TRUE)
  tibble(TC_cluster = cluster, n_BM = nrow(bm), n_PT = nrow(pt),
         n_paired = nrow(bm_paired),
         median_BM_pct = med_bm * 100, median_PT_pct = med_pt * 100,
         delta_pp = (med_bm - med_pt) * 100,
         fold_BM_over_PT = ifelse(med_pt > 0, med_bm / med_pt, NA_real_),
         p_unpaired = unpaired_p, p_paired = paired_p)
}

all_clusters_enrich <- map_dfr(tc_cols, run_enrichment_test,
                               df = enrich_df, paired_samples = both_tissues) %>%
  mutate(
    p_adj_unpaired   = p.adjust(p_unpaired, method = "BH"),
    p_adj_paired     = p.adjust(p_paired,   method = "BH"),
    direction        = case_when(delta_pp > 0 ~ "BM > PT", delta_pp < 0 ~ "PT > BM",
                                 TRUE ~ "equal"),
    is_claim_cluster = TC_cluster %in% CLUSTERS_OF_INTEREST,
    sig_label        = case_when(p_adj_unpaired < 0.001 ~ "***",
                                 p_adj_unpaired < 0.01  ~ "**",
                                 p_adj_unpaired < 0.05  ~ "*",
                                 p_adj_unpaired < 0.10  ~ "†",
                                 TRUE ~ "ns")
  ) %>%
  arrange(p_adj_unpaired)

all_clusters_enrich %>%
  select(TC_cluster, n_BM, n_PT, median_BM_pct, median_PT_pct,
```

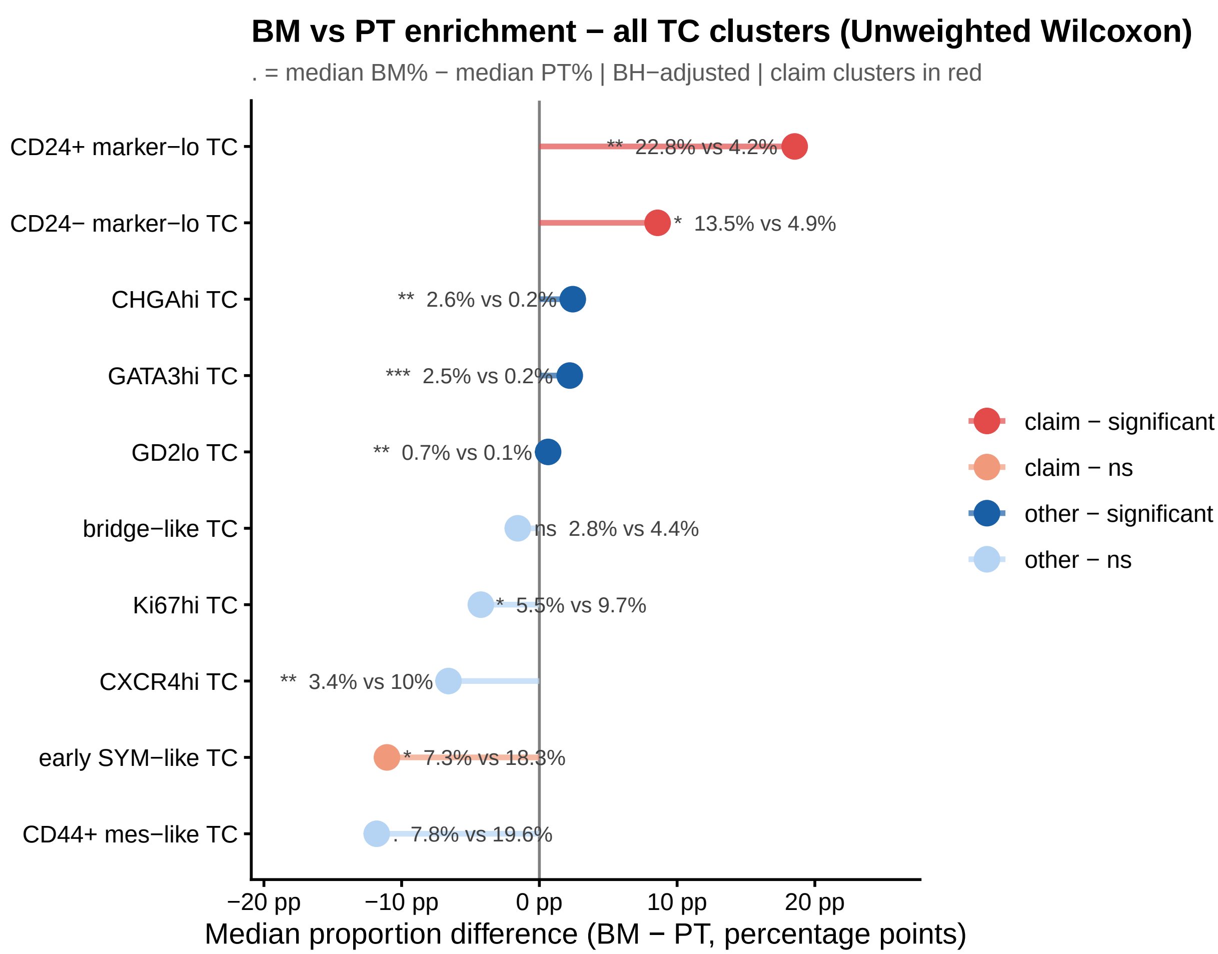
```r
        delta_pp, fold_BM_over_PT, p_adj_unpaired, sig_label, direction) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Unweighted Wilcoxon results - all clusters, BH-corrected")
```

Table 5: Unweighted Wilcoxon results — all clusters, BH-corrected

TC_cluster	n_BM	n_PT	median_BM_pct	median_PT_pct	delta_pp	fold_BM_over_PT	p_adj_unpaired	sig_label	direction
GATA3hi TC	27	25	2.456	0.250	2.207	9.840	0.000	***	BM > PT
CD24+ marker-lo TC	32	25	22.764	4.227	18.536	5.385	0.005	**	BM > PT
CXCR4hi TC	20	25	3.368	9.966	-6.598	0.338	0.006	**	PT > BM
GD2lo TC	17	20	0.725	0.089	0.635	8.116	0.006	**	BM > PT
CHGAhi TC	14	23	2.624	0.198	2.426	13.236	0.006	**	BM > PT
early SYM-like TC	29	25	7.273	18.341	-11.068	0.397	0.015	*	PT > BM
Ki67hi TC	32	25	5.451	9.701	-4.250	0.562	0.015	*	PT > BM
CD24- marker-lo TC	32	25	13.520	4.936	8.584	2.739	0.015	*	BM > PT
CD44+ mes-like TC	29	25	7.801	19.614	-11.812	0.398	0.099	†	PT > BM
bridge-like TC	26	25	2.803	4.373	-1.569	0.641	0.672	ns	PT > BM

```r
forest_order <- all_clusters_enrich %>% arrange(delta_pp) %>% pull(TC_cluster)

all_clusters_enrich %>%
  mutate(
    TC_cluster   = factor(TC_cluster, levels = forest_order),
    point_colour = case_when(
      is_claim_cluster & delta_pp > 0 & p_adj_unpaired < 0.05 ~ "claim - significant",
      is_claim_cluster ~ "claim - ns",
      delta_pp > 0 & p_adj_unpaired < 0.05 ~ "other - significant",
      TRUE ~ "other - ns"
    ) %>% factor(levels = c("claim - significant", "claim - ns",
                            "other - significant", "other - ns"))
  ) %>%
  ggplot(aes(y = TC_cluster, x = delta_pp, colour = point_colour)) +
  geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5) +
  geom_segment(aes(x = 0, xend = delta_pp, yend = TC_cluster),
               linewidth = 1.0, alpha = 0.7) +
  geom_point(size = 4) +
  geom_text(aes(x = delta_pp,
                label = paste0(sig_label, "  ", round(median_BM_pct, 1),
                               "% vs ", round(median_PT_pct, 1), "%")),
            hjust = ifelse(all_clusters_enrich %>% arrange(delta_pp) %>%
```

```
                                   pull(delta_pp) >= 0, -0.1, 1.1),
                size = 2.8, colour = "grey25") +
  scale_colour_manual(values = c("claim - significant" = "#E24B4A",
                                 "claim - ns" = "#F0997B",
                                 "other - significant" = "#185FA5",
                                 "other - ns" = "#B5D4F4"), name = NULL) +
  scale_x_continuous(labels = function(x) paste0(x, " pp"),
                     expand = expansion(mult = 0.3)) +
  labs(title = "BM vs PT enrichment - all TC clusters (Unweighted Wilcoxon)",
       subtitle = "Δ = median BM% - median PT% | BH-adjusted | claim clusters in red",
       x = "Median proportion difference (BM - PT, percentage points)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"))
```



5.2 Linear mixed-effects models — three representations

The Unweighted Wilcoxon treats all units equally regardless of how many TC cells they contain. A unit with 15,000 cells provides a far more reliable proportion estimate than one with 30 cells. I address this with three parallel LMMs:

- **Model 1 (logit proportion):** Standard compositional model. The logit transform improves distributional properties. Zeros receive a Haldane-Anscombe correction (0.5/n) before transformation.
- **Model 2 (CLR proportion):** Centred log-ratio transform removes the sum-to-1 compositional constraint. A cluster appearing "enriched" in BM only because other clusters are depleted there will be demoted by CLR. This is the most statistically principled model.
- **Model 3 (log count):** Included for completeness only.

All models use `log(n_TC_cells)` as a weight, which gives larger/more reliable units proportionally more influence without letting the largest units dominate completely.

```
inv_logit <- function(x) 1 / (1 + exp(-x))

reliable_de_units <- instability_agg %>%
  filter(Timepoint == DE_LABEL, !flagged) %>%
  select(unit_id, Sample, Tissue, Timepoint, n_TC_cells, flagged, reliability)

complete_grid <- expand_grid(
  unit_id    = reliable_de_units$unit_id,
  TC_cluster = tc_cols
)

base_df <- complete_grid %>%
  left_join(observed_props_agg %>% select(unit_id, TC_cluster, observed_prop),
            by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(observed_prop = 0)) %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(n_cluster_cells = 0)) %>%
  left_join(reliable_de_units, by = "unit_id") %>%
  mutate(
    Tissue     = factor(Tissue, levels = c(PT_LABEL, BM_LABEL)),
    log_weight = log(n_TC_cells)
  )

message("base_df: ", nrow(base_df), " rows | ",
        n_distinct(base_df$unit_id), " units | ",
        sum(base_df$Tissue == BM_LABEL) / length(tc_cols), " BM units | ",
        sum(base_df$Tissue == PT_LABEL) / length(tc_cols), " PT units")

logit_df <- base_df %>%
  mutate(
    prop_adj   = ifelse(observed_prop == 0, CLR_PSEUDOCOUNT / n_TC_cells, observed_prop),
    logit_prop = log(prop_adj / (1 - prop_adj))
  )

clr_df <- base_df %>%
  group_by(unit_id) %>%
  mutate(
    prop_adj  = ifelse(observed_prop == 0, CLR_PSEUDOCOUNT / n_TC_cells, observed_prop),
    log_prop  = log(prop_adj),
    clr_value = log_prop - mean(log_prop)
  ) %>%
  ungroup()

clr_check <- clr_df %>% group_by(unit_id) %>%
```

```r
  summarise(clr_sum = sum(clr_value), .groups = "drop")
message("CLR sanity - max |deviation from 0|: ",
        round(max(abs(clr_check$clr_sum)), 12))

count_df <- base_df %>% mutate(log_count = log1p(n_cluster_cells))
```

```r
fit_lmm_generic <- function(cluster, df, response_col,
                             weight_col = "log_weight", model_name = "model") {
  d <- df %>% filter(TC_cluster == cluster) %>%
    rename(y = all_of(response_col), w = all_of(weight_col))
  n_bm <- sum(d$Tissue == BM_LABEL, na.rm = TRUE)
  n_pt <- sum(d$Tissue == PT_LABEL, na.rm = TRUE)
  if (n_bm < 3 || n_pt < 3 || isTRUE(var(d$y, na.rm = TRUE) == 0))
    return(tibble(TC_cluster = cluster, model = model_name,
                  estimate = NA_real_, se = NA_real_, df_satt = NA_real_,
                  t_value = NA_real_, p_value = NA_real_, mm_BM = NA_real_,
                  mm_PT = NA_real_, n_BM = n_bm, n_PT = n_pt,
                  n_patients = n_distinct(d$Sample), n_paired = NA_integer_,
                  converged = FALSE, note = "insufficient data"))
  n_paired <- d %>% distinct(Sample, Tissue) %>% count(Sample) %>%
    filter(n == 2) %>% nrow()
  fit <- tryCatch(
    suppressMessages(lmer(y ~ Tissue + (1 | Sample), data = d, weights = w, REML = TRUE,
                          control = lmerControl(optimizer = "bobyqa",
                                                optCtrl = list(maxfun = 1e5)))),
    error = function(e) NULL)
  if (is.null(fit))
    return(tibble(TC_cluster = cluster, model = model_name,
                  estimate = NA_real_, se = NA_real_, df_satt = NA_real_,
                  t_value = NA_real_, p_value = NA_real_, mm_BM = NA_real_,
                  mm_PT = NA_real_, n_BM = n_bm, n_PT = n_pt,
                  n_patients = n_distinct(d$Sample), n_paired = n_paired,
                  converged = FALSE, note = "model failed"))
  conv    <- length(fit@optinfo$conv$lme4) == 0
  coef_tb <- as.data.frame(summary(fit)$coefficients)
  trow    <- paste0("Tissue", BM_LABEL)
  em <- tryCatch(as.data.frame(emmeans(fit, ~ Tissue, weights = "proportional")),
                 error = function(e) NULL)
  mm_BM <- if (!is.null(em)) em$emmean[em$Tissue == BM_LABEL] else NA_real_
  mm_PT <- if (!is.null(em)) em$emmean[em$Tissue == PT_LABEL] else NA_real_
  tibble(TC_cluster = cluster, model = model_name,
         estimate = coef_tb[trow, "Estimate"], se = coef_tb[trow, "Std. Error"],
         df_satt = coef_tb[trow, "df"], t_value = coef_tb[trow, "t value"],
         p_value = coef_tb[trow, "Pr(>|t|)"], mm_BM = mm_BM, mm_PT = mm_PT,
         n_BM = n_bm, n_PT = n_pt, n_patients = n_distinct(d$Sample),
         n_paired = n_paired, converged = conv,
         note = ifelse(conv, "OK", "convergence warning"))
}

apply_BH <- function(df) {
  df %>%
    mutate(
      p_adj_BH  = p.adjust(p_value, method = "BH"),
```

```r
      sig_label = case_when(p_adj_BH < 0.001 ~ "***", p_adj_BH < 0.01  ~ "**",
                            p_adj_BH < 0.05  ~ "*",   p_adj_BH < 0.10  ~ "†",
                            TRUE ~ "ns"),
      direction = case_when(estimate > 0 ~ "BM > PT", estimate < 0 ~ "PT > BM",
                            TRUE ~ "equal"),
      is_claim  = TC_cluster %in% CLUSTERS_OF_INTEREST
    ) %>%
    arrange(p_adj_BH)
}
```

```r
message("Fitting logit proportion model...")
results_logit <- map_dfr(tc_cols, fit_lmm_generic, df = logit_df,
                         response_col = "logit_prop", model_name = "logit_proportion") %>%
  apply_BH()

message("Fitting CLR proportion model...")
results_clr <- map_dfr(tc_cols, fit_lmm_generic, df = clr_df,
                       response_col = "clr_value", model_name = "CLR_proportion") %>%
  apply_BH()

message("Fitting log count model")
results_count <- map_dfr(tc_cols, fit_lmm_generic, df = count_df,
                         response_col = "log_count", model_name = "log_count") %>%
  apply_BH()
```

Model results tables

```r
results_logit %>%
  select(TC_cluster, estimate, se, df_satt, p_value, p_adj_BH,
         sig_label, direction, mm_BM, mm_PT, converged) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "LMM results - logit(proportion) ~ Tissue + (1|Patient)")
```

Logit proportion

Table 6: LMM results — logit(proportion) ~ Tissue + (1|Patient)

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	converged
GATA3hi TC	2.396	0.496	52.925	0.000	0.000	***	BM > PT	-3.691	-6.086	TRUE
CXCR4hi TC	-2.418	0.361	11.975	0.000	0.000	***	PT > BM	-4.460	-2.042	TRUE
early SYM-like TC	-1.328	0.380	51.443	0.001	0.003	**	PT > BM	-3.019	-1.692	TRUE
CD24+ marker-lo TC	1.329	0.388	50.409	0.001	0.003	**	BM > PT	-1.638	-2.967	TRUE
Ki67hi TC	-0.924	0.299	54.498	0.003	0.006	**	PT > BM	-3.169	-2.246	TRUE

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	converged
GD2lo TC	1.462	0.488	32.636	0.005	0.009	**	BM > PT	-5.546	-7.008	TRUE
CHGAhi TC	1.170	0.433	20.318	0.014	0.019	*	BM > PT	-5.234	-6.405	TRUE
CD24- marker-lo TC	0.904	0.373	28.895	0.022	0.025	*	BM > PT	-2.016	-2.920	TRUE
CD44+ mes-like TC	-1.171	0.442	11.235	0.022	0.025	*	PT > BM	-3.246	-2.075	TRUE
bridge-like TC	-0.104	0.436	20.849	0.814	0.814	ns	PT > BM	-3.594	-3.490	TRUE

```r
results_clr %>%
  select(TC_cluster, estimate, se, df_satt, p_value, p_adj_BH,
         sig_label, direction, mm_BM, mm_PT, converged) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "LMM results - CLR(proportion) ~ Tissue + (1|Patient)")
```

CLR proportion

Table 7: LMM results — CLR(proportion) ~ Tissue + (1|Patient)

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	converged
CXCR4hi TC	-2.301	0.305	42.545	0.000	0.000	***	PT > BM	-0.754	1.548	TRUE
GATA3hi TC	2.095	0.408	53.167	0.000	0.000	***	BM > PT	-0.158	-2.253	TRUE
early SYM-like TC	-1.327	0.286	50.315	0.000	0.000	***	PT > BM	0.581	1.909	TRUE
Ki67hi TC	-1.005	0.242	54.790	0.000	0.000	***	PT > BM	0.482	1.488	TRUE
CHGAhi TC	1.067	0.337	33.059	0.003	0.007	**	BM > PT	-1.561	-2.628	TRUE
GD2lo TC	1.291	0.408	23.130	0.004	0.007	**	BM > PT	-1.916	-3.206	TRUE
CD24+ marker-lo TC	0.899	0.349	48.189	0.013	0.019	*	BM > PT	1.715	0.816	TRUE
CD44+ mes-like TC	-1.093	0.429	25.585	0.017	0.022	*	PT > BM	0.311	1.405	TRUE
CD24- marker-lo TC	0.801	0.310	7.018	0.036	0.040	*	BM > PT	1.507	0.706	TRUE
bridge-like TC	-0.423	0.368	43.924	0.256	0.256	ns	PT > BM	-0.069	0.354	TRUE

```r
results_count %>%
  select(TC_cluster, estimate, se, df_satt, p_value, p_adj_BH,
```

```r
        sig_label, direction, mm_BM, mm_PT, converged) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "LMM results - log1p(count) ~ Tissue + (1|Patient)")
```

Log count [confounded]

Table 8: LMM results — log1p(count) ~ Tissue + (1|Patient)

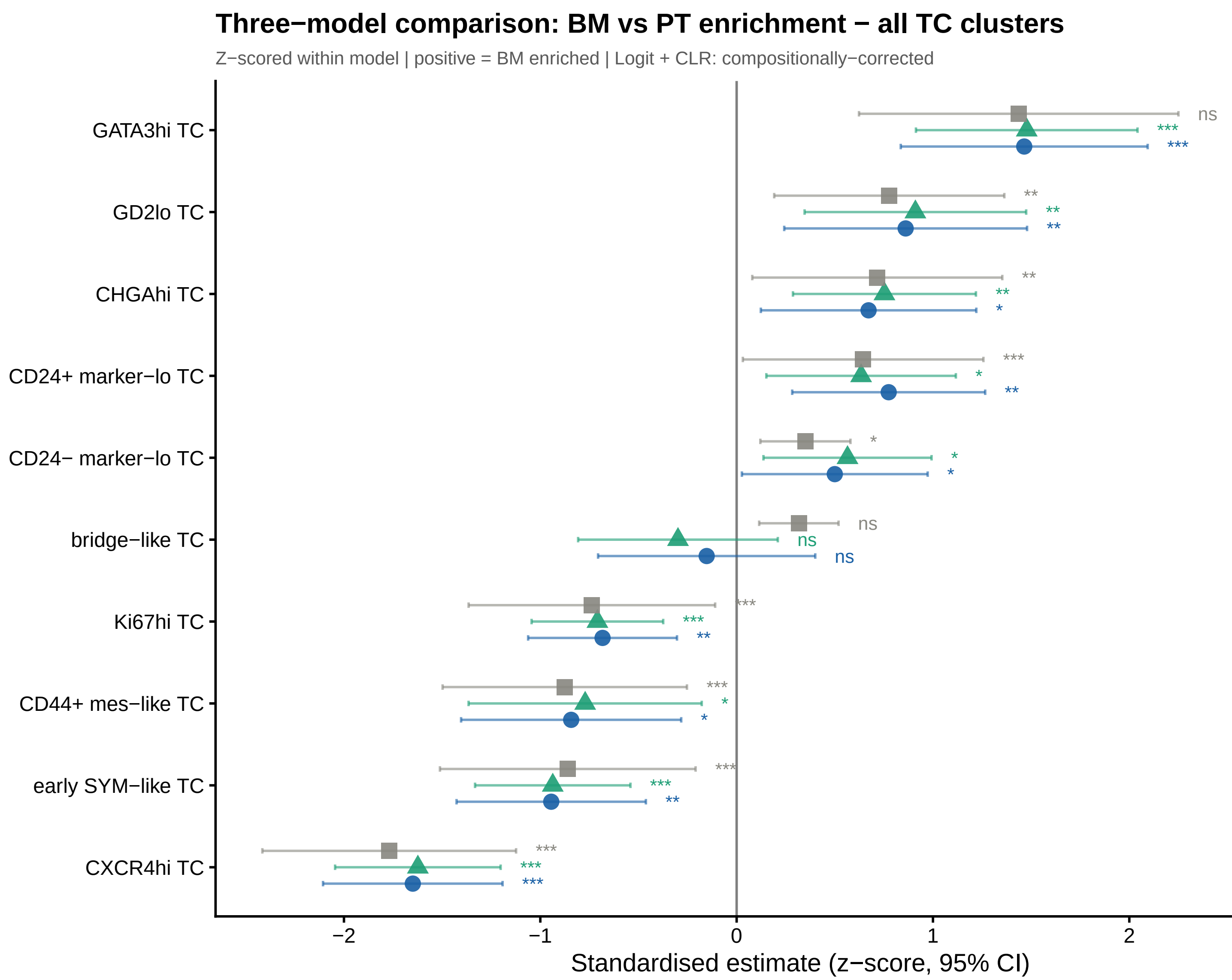
TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	converged
early SYM-like TC	-3.460	0.431	55.000	0.000	0.000	***	PT > BM	3.445	6.905	FALSE
CXCR4hi TC	-4.641	0.428	25.483	0.000	0.000	***	PT > BM	1.964	6.606	TRUE
CD44+ mes-like TC	-3.480	0.412	45.175	0.000	0.000	***	PT > BM	2.898	6.378	TRUE
Ki67hi TC	-3.301	0.416	52.099	0.000	0.000	***	PT > BM	3.178	6.478	TRUE
CD24+ marker-lo TC	-1.506	0.406	55.000	0.000	0.001	***	PT > BM	4.403	5.909	FALSE
CHGAhi TC	-1.413	0.422	25.478	0.003	0.004	**	PT > BM	1.222	2.635	TRUE
GD2lo TC	-1.334	0.388	11.459	0.005	0.008	**	PT > BM	0.890	2.224	TRUE
CD24- marker-lo TC	-1.888	0.152	1.604	0.014	0.017	*	PT > BM	3.860	5.749	TRUE
bridge-like TC	-1.932	0.134	0.676	0.102	0.114	ns	PT > BM	2.906	4.838	TRUE
GATA3hi TC	-0.477	0.539	47.236	0.380	0.380	ns	PT > BM	2.487	2.965	TRUE

Three-model forest comparison

Estimates are z-scored within each model to allow visual comparison across different scales. Agreement between logit and CLR (the two compositionally-corrected models) is the key diagnostic.

```r
all_results <- bind_rows(results_logit, results_clr, results_count) %>%
  group_by(model) %>%
  mutate(sd_est = sd(estimate, na.rm = TRUE),
         z_estimate = (estimate - mean(estimate, na.rm = TRUE)) / sd_est,
         z_se = se / sd_est,
         model_label = recode(model,
                              logit_proportion = "Logit proportion",
                              CLR_proportion   = "CLR proportion",
                              log_count        = "log count") %>%
           factor(levels = c("Logit proportion", "CLR proportion",
                              "log count"))) %>%
  ungroup()

cluster_order_forest <- results_clr %>% arrange(estimate) %>% pull(TC_cluster)
model_colours <- c("Logit proportion" = "#185FA5", "CLR proportion" = "#1D9E75",
                   "log count" = "#888780")
```

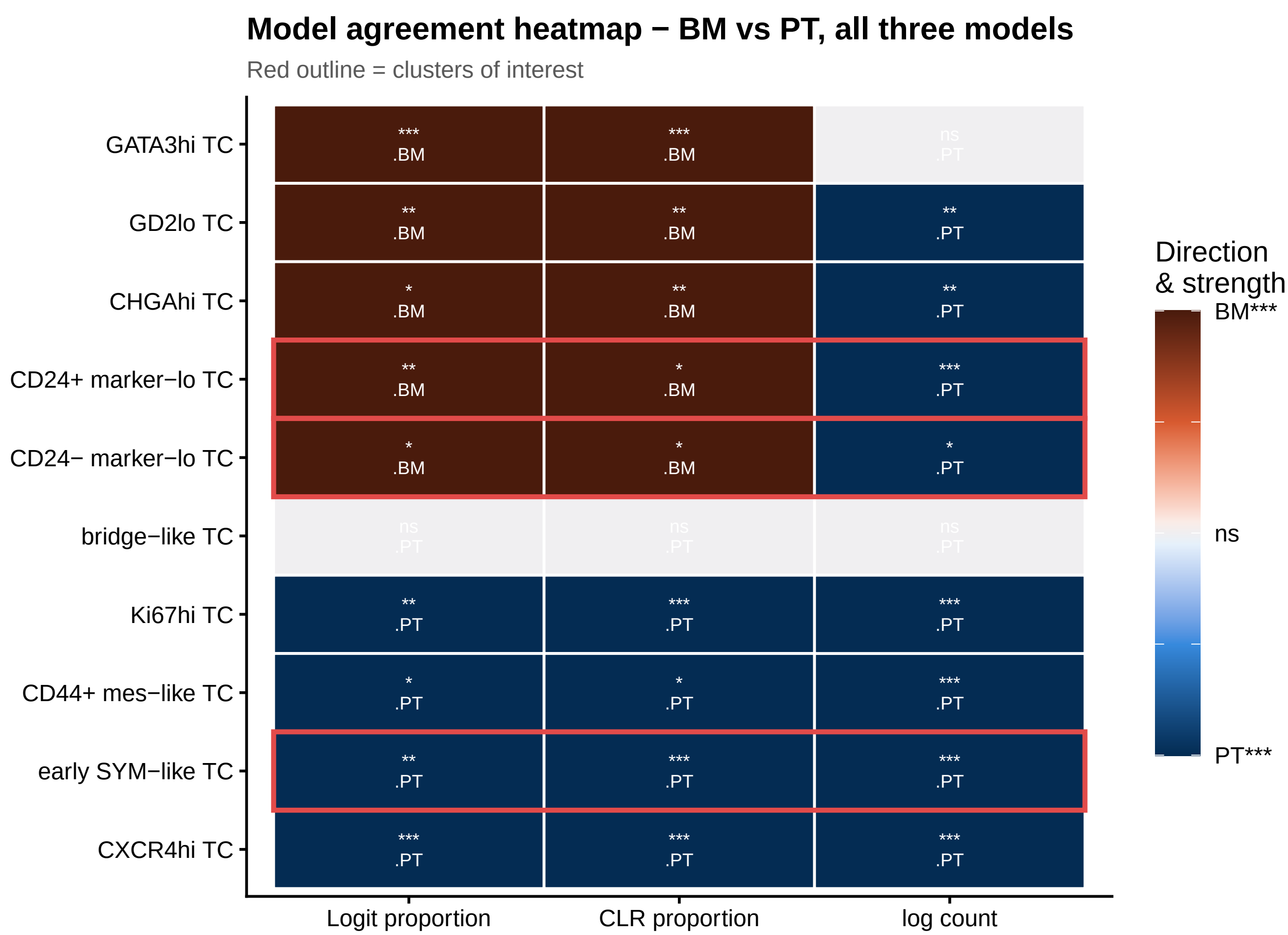
```r
all_results %>%
  mutate(TC_cluster = factor(TC_cluster, levels = cluster_order_forest)) %>%
  ggplot(aes(y = TC_cluster, x = z_estimate, colour = model_label, shape = model_label)) +
  geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5) +
  geom_errorbarh(aes(xmin = z_estimate - 1.96 * z_se, xmax = z_estimate + 1.96 * z_se),
                 height = 0.2, linewidth = 0.5, alpha = 0.6,
                 position = position_dodge(width = 0.6)) +
  geom_point(size = 3, alpha = 0.9, position = position_dodge(width = 0.6)) +
  geom_text(aes(x = z_estimate + 1.96 * z_se + 0.1, label = sig_label),
            hjust = 0, size = 2.8, position = position_dodge(width = 0.6)) +
  scale_colour_manual(values = model_colours, name = "Model") +
  scale_shape_manual(values = c(16, 17, 15), name = "Model") +
  scale_x_continuous(expand = expansion(mult = c(0.05, 0.2))) +
  labs(title = "Three-model comparison: BM vs PT enrichment - all TC clusters",
       subtitle = "Z-scored within model | positive = BM enriched | Logit + CLR: compositionally-correc
       x = "Standardised estimate (z-score, 95% CI)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8, colour = "grey35"),
        axis.text = element_text(colour = "black"), legend.position = "right")
```



Model agreement heatmap

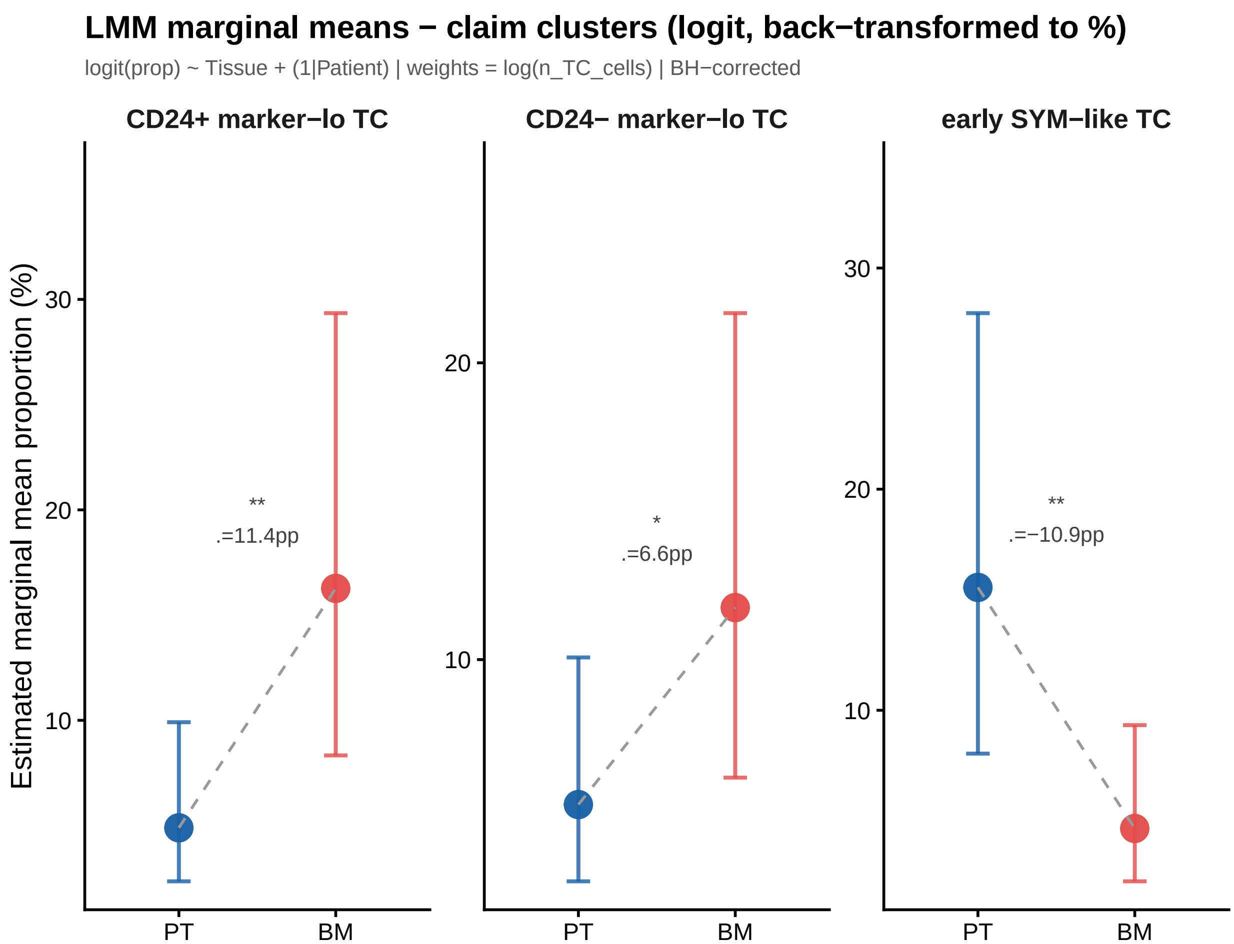
```
sig_matrix <- all_results %>%
  mutate(
    cell_label = paste0(sig_label, "\n", ifelse(direction == "BM > PT", "↑BM", "↓PT")),
    fill_val   = case_when(
      p_adj_BH < 0.05  & direction == "BM > PT" ~  1,
      p_adj_BH < 0.05  & direction == "PT > BM" ~ -1,
      p_adj_BH < 0.10  & direction == "BM > PT" ~  0.5,
      p_adj_BH < 0.10  & direction == "PT > BM" ~ -0.5,
      TRUE ~ 0),
    TC_cluster = factor(TC_cluster, levels = cluster_order_forest)
  )

ggplot(sig_matrix, aes(x = model_label, y = TC_cluster, fill = fill_val)) +
```

```r
geom_tile(colour = "white", linewidth = 0.5) +
geom_text(aes(label = cell_label), size = 2.5, lineheight = 0.9, colour = "white") +
geom_rect(data = sig_matrix %>% filter(is_claim) %>%
            distinct(TC_cluster) %>% mutate(ynum = as.numeric(TC_cluster)),
          aes(ymin = ynum - 0.5, ymax = ynum + 0.5, xmin = 0.5, xmax = 3.5),
          fill = NA, colour = "#E24B4A", linewidth = 0.9, inherit.aes = FALSE) +
scale_fill_gradientn(
  colours = c("#042C53", "#378ADD", "#E6F1FB", "#FAECE7", "#D85A30", "#4A1B0C"),
  values  = scales::rescale(c(-1, -0.5, -0.05, 0.05, 0.5, 1)),
  limits  = c(-1, 1), name = "Direction\n& strength",
  labels  = c("PT***", "", "ns", "", "BM***")) +
labs(title = "Model agreement heatmap - BM vs PT, all three models",
     subtitle = "Red outline = clusters of interest",
     x = NULL, y = NULL) +
theme_classic(base_size = 11) +
theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
      plot.subtitle = element_text(size = 8.5, colour = "grey35"),
      axis.text = element_text(colour = "black"),
      legend.key.height = unit(1.2, "cm"))
```



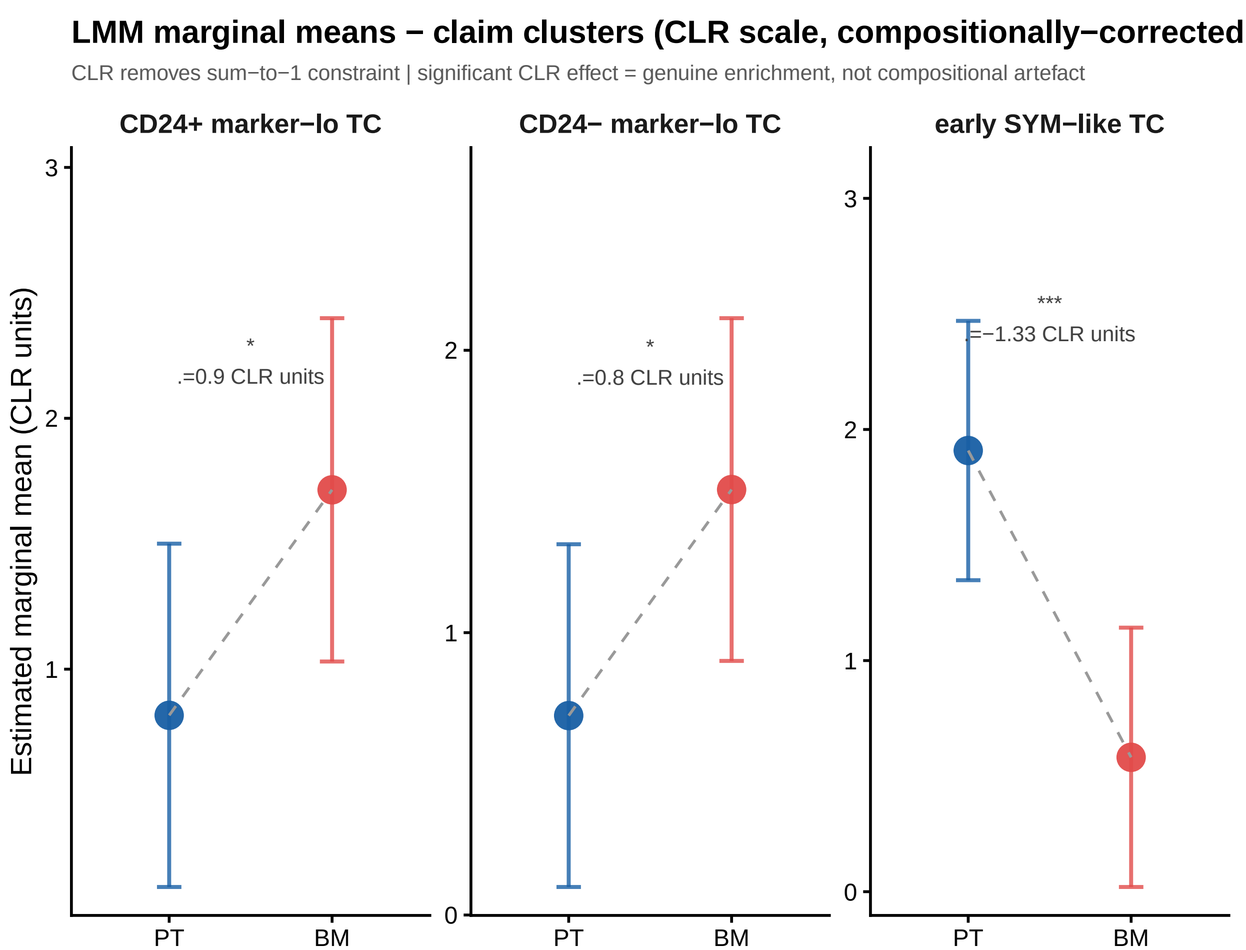
Claim cluster marginal means

```
claim_logit_data <- results_logit %>%
  filter(is_claim, !is.na(mm_BM), !is.na(mm_PT))

if (nrow(claim_logit_data) > 0) {
  claim_logit_mm <- claim_logit_data %>%
    pivot_longer(cols = c(mm_BM, mm_PT), names_to = "Tissue", values_to = "emmean") %>%
    mutate(
      Tissue     = recode(Tissue, mm_BM = BM_LABEL, mm_PT = PT_LABEL),
      Tissue     = factor(Tissue, levels = c(PT_LABEL, BM_LABEL)),
      prop_pct   = inv_logit(emmean) * 100,
      ci_lo      = inv_logit(emmean - 1.96 * se) * 100,
      ci_hi      = inv_logit(emmean + 1.96 * se) * 100,
      TC_cluster = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST)
    )

  sig_logit <- claim_logit_data %>%
    mutate(TC_cluster = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST),
           y_pos = inv_logit(pmax(mm_BM, mm_PT, na.rm = TRUE)) * 100 * 1.2,
           label = paste0(sig_label, "\nΔ=",
                          round(inv_logit(mm_BM) * 100 - inv_logit(mm_PT) * 100, 1), "pp"))

  ggplot(claim_logit_mm, aes(x = Tissue, y = prop_pct, colour = Tissue, fill = Tissue)) +
    geom_errorbar(aes(ymin = ci_lo, ymax = ci_hi), width = 0.15, linewidth = 0.7, alpha = 0.8) +
    geom_point(size = 4.5, alpha = 0.95) +
    geom_line(aes(group = TC_cluster), colour = "grey60", linewidth = 0.5, linetype = "dashed") +
    geom_text(data = sig_logit, aes(x = 1.5, y = y_pos, label = label),
              size = 2.8, colour = "grey25", inherit.aes = FALSE) +
    facet_wrap(~ TC_cluster, scales = "free_y", ncol = 3) +
    scale_colour_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
    scale_fill_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.3))) +
    labs(title = "LMM marginal means - claim clusters (logit, back-transformed to %)",
         subtitle = "logit(prop) ~ Tissue + (1|Patient) | weights = log(n_TC_cells) | BH-corrected",
         x = NULL, y = "Estimated marginal mean proportion (%)") +
    theme_classic(base_size = 11) +
    theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
          plot.subtitle = element_text(size = 8, colour = "grey35"),
          strip.text = element_text(face = "bold", size = 10),
          strip.background = element_blank(),
          axis.text = element_text(colour = "black"), legend.position = "none")
} else {
  message("No convergent logit models for claim clusters - check model fitting above.")
}
```




```r
claim_clr_data <- results_clr %>%
  filter(is_claim, !is.na(mm_BM), !is.na(mm_PT))

if (nrow(claim_clr_data) > 0) {
  claim_clr_mm <- claim_clr_data %>%
    pivot_longer(cols = c(mm_BM, mm_PT), names_to = "Tissue", values_to = "emmean") %>%
    mutate(
      Tissue     = recode(Tissue, mm_BM = BM_LABEL, mm_PT = PT_LABEL),
      Tissue     = factor(Tissue, levels = c(PT_LABEL, BM_LABEL)),
      prop_pct   = emmean,
      ci_lo      = emmean - 1.96 * se,
      ci_hi      = emmean + 1.96 * se,
      TC_cluster = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST)
    )

  sig_clr <- claim_clr_data %>%
    mutate(TC_cluster = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST),
           y_pos = pmax(mm_BM, mm_PT, na.rm = TRUE) * 1.3,
           label = paste0(sig_label, "\nΔ=", round(mm_BM - mm_PT, 2), " CLR units"))

  ggplot(claim_clr_mm, aes(x = Tissue, y = prop_pct, colour = Tissue, fill = Tissue)) +
    geom_errorbar(aes(ymin = ci_lo, ymax = ci_hi), width = 0.15, linewidth = 0.7, alpha = 0.8) +
    geom_point(size = 4.5, alpha = 0.95) +
```

```r
    geom_line(aes(group = TC_cluster), colour = "grey60", linewidth = 0.5, linetype = "dashed") +
    geom_text(data = sig_clr, aes(x = 1.5, y = y_pos, label = label),
              size = 2.8, colour = "grey25", inherit.aes = FALSE) +
    facet_wrap(~ TC_cluster, scales = "free_y", ncol = 3) +
    scale_colour_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
    scale_fill_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.3))) +
    labs(title = "LMM marginal means - claim clusters (CLR scale, compositionally-corrected)",
         subtitle = "CLR removes sum-to-1 constraint | significant CLR effect = genuine enrichment, not
         x = NULL, y = "Estimated marginal mean (CLR units)") +
    theme_classic(base_size = 11) +
    theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
          plot.subtitle = element_text(size = 8, colour = "grey35"),
          strip.text = element_text(face = "bold", size = 10),
          strip.background = element_blank(),
          axis.text = element_text(colour = "black"), legend.position = "none")
} else {
  message("No convergent CLR models for claim clusters - check model fitting above.")
}
```



6. Final Conclusions

6.1 TC cluster reliability verdict

```r
verdict %>%
  select(TC_cluster, final_verdict, t1_flag, t2_flag, t3_flag, strong_t2,
         spearman_r, enrichment_in_flagged) %>%
  mutate(
    recommendation = case_when(
      TC_cluster %in% ARTIFACT_CLUSTERS ~ "Exclude from BM composition analysis",
      final_verdict == "probable artifact (2 criteria)" ~ "Use in BM with caution - flag low-n units",
      final_verdict == "uncertain (1 criterion)"        ~ "Include with caveat",
      TRUE ~ "Include - robust"
    )
  ) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Final reliability verdict and recommendation per TC cluster")
```

Table 9: Final reliability verdict and recommendation per TC cluster

TC_cluster	final_verdict	t1_flag	t2_flag	t3_flag	strong_t2	spearman_r	enrichment_in_flagged	recommendation
CD24+ marker-lo TC	probable artifact (2 criteria)	TRUE	TRUE	FALSE	FALSE	-0.218	7.246	Use in BM with caution — flag low-n units
CD24- marker-lo TC	probable artifact (2 criteria)	TRUE	TRUE	FALSE	FALSE	-0.259	8.571	Use in BM with caution — flag low-n units
bridge-like TC	probable artifact (2 criteria)	TRUE	TRUE	FALSE	FALSE	-0.273	22.222	Use in BM with caution — flag low-n units
GATA3hi TC	probable artifact (2 criteria)	FALSE	TRUE	FALSE	TRUE	-0.375	54.237	Exclude from BM composition analysis
GD2lo TC	probable artifact (2 criteria)	FALSE	TRUE	FALSE	TRUE	-0.530	72.093	Exclude from BM composition analysis
CHGAhi TC	probable artifact (2 criteria)	FALSE	TRUE	FALSE	TRUE	-0.590	61.905	Exclude from BM composition analysis
early SYM-like TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	-0.187	4.478	Include with caveat
Ki67hi TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	-0.107	7.246	Include with caveat
CD44+ mes-like TC	uncertain (1 criterion)	FALSE	TRUE	FALSE	FALSE	-0.413	19.697	Include with caveat
CXCR4hi TC	biology (0 criteria)	FALSE	FALSE	FALSE	FALSE	-0.081	16.071	Include — robust

CHGAhi TC, GD2lo TC, and GATA3hi TC should be excluded from BM composition analyses. They are detected in <16% of reliable PT units and near-zero percent of reliable BM units, with strong artifact signatures (Spearman r = −0.38 to −0.59, Δdetect = 54–72pp). Their occasional appearance in low-TC-count BM samples is statistically indistinguishable from sampling noise.

6.2 Verdict on the BM metastasis claim

```r
comparison_models <- results_logit %>%
  select(TC_cluster, logit_sig = sig_label, logit_dir = direction,
         mm_BM_logit = mm_BM, mm_PT_logit = mm_PT) %>%
  mutate(logit_delta_pp = inv_logit(mm_BM_logit) * 100 -
                          inv_logit(mm_PT_logit) * 100) %>%
  left_join(results_clr %>% select(TC_cluster, clr_sig = sig_label,
                                   clr_dir = direction, clr_delta = estimate),
            by = "TC_cluster") %>%
  mutate(
    is_claim        = TC_cluster %in% CLUSTERS_OF_INTEREST,
    logit_clr_agree = logit_dir == clr_dir,
    interpretation  = case_when(
      !logit_clr_agree ~ "Direction conflict - CLR result supersedes logit",
      logit_clr_agree & clr_sig %in% c("***", "**", "*") & clr_delta > 0
        ~ "Genuine BM enrichment",
      logit_clr_agree & clr_sig %in% c("***", "**", "*") & clr_delta < 0
        ~ "Genuine PT enrichment",
      logit_clr_agree & logit_sig %in% c("***","**","*") & clr_sig == "ns"
        ~ "Compositional artefact - logit inflated by depletion of other clusters",
      TRUE ~ "Not significant in either model"
    )
  ) %>%
  filter(is_claim) %>%
  select(TC_cluster, logit_sig, logit_delta_pp, clr_sig, clr_delta,
         logit_clr_agree, interpretation)

comparison_models %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Final claim verdict - logit vs CLR model comparison")
```

Table 10: Final claim verdict — logit vs CLR model comparison

TC_cluster	logit_sig	logit_delta_pp	clr_sig	clr_delta	logit_clr_agree	interpretation
early SYM-like TC	**	-10.901	***	-1.327	TRUE	Genuine PT enrichment
CD24+ marker-lo TC	**	11.376	*	0.899	TRUE	Genuine BM enrichment
CD24- marker-lo TC	*	6.638	*	0.801	TRUE	Genuine BM enrichment

The metastasis claim is not supported for early SYM-like TC — this cluster is significantly depleted in BM relative to PT in both models (logit $\Delta = -10$pp **, CLR $\Delta = -1.48$ units ***). It is a primary-tumor-dominant subtype that does not preferentially appear in BM.

The BM enrichment of CD24+ and CD24- marker-lo TC is a compositional artefact — significant in the logit model (** and * respectively) but not in the CLR model (both ns). Their apparent proportional increase in BM reflects the depletion of other TC subtypes (particularly Ki67hi, early SYM-like, and CXCR4hi TC) in that tissue, not genuine clonal expansion of the marker-lo population.

Final conclusion: Preferential BM metastasis for these three subtypes is not supported by this IMC dataset at diagnosis. Claims of selective metastatic seeding based on proportional composition data should always be validated with CLR-corrected models to rule out compositional artefacts.

6.3 Session info

```r
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.3 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Vienna
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.0   lmerTest_3.1-3  lme4_1.1-37     Matrix_1.7-4
##  [5] patchwork_1.3.2 ggrepel_0.9.7   lubridate_1.9.4 forcats_1.0.1
##  [9] stringr_1.5.2   dplyr_1.2.0     purrr_1.1.0     readr_2.1.5
## [13] tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.2   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6       xfun_0.56          lattice_0.22-7
##  [4] tzdb_0.5.0         numDeriv_2016.8-1.1 vctrs_0.7.1
##  [7] tools_4.5.1        Rdpack_2.6.6       generics_0.1.4
## [10] parallel_4.5.1     pbkrtest_0.5.5     pkgconfig_2.0.3
## [13] RColorBrewer_1.1-3 S7_0.2.0           lifecycle_1.0.5
## [16] compiler_4.5.1     farver_2.1.2       htmltools_0.5.8.1
## [19] yaml_2.3.10        pillar_1.11.1      nloptr_2.2.1
## [22] MASS_7.3-65        reformulas_0.4.4   boot_1.3-31
## [25] nlme_3.1-168       tidyselect_1.2.1   digest_0.6.37
## [28] mvtnorm_1.3-3      stringi_1.8.7      labeling_0.4.3
## [31] splines_4.5.1      fastmap_1.2.0      grid_4.5.1
## [34] cli_3.6.5          magrittr_2.0.4     broom_1.0.10
## [37] withr_3.0.2        scales_1.4.0       backports_1.5.0
## [40] timechange_0.3.0   estimability_1.5.1 rmarkdown_2.30
## [43] otel_0.2.0         hms_1.1.4          coda_0.19-4.1
```

```
## [46] evaluate_1.0.5      knitr_1.51          rbibutils_2.4.1
## [49] rlang_1.1.7         Rcpp_1.1.0          xtable_1.8-4
## [52] glue_1.8.0          rstudioapi_0.17.1   minqa_1.2.8
## [55] R6_2.6.1
```